

PROBLEMS 20.5

1. Evaluate $\int_0^{1+i} (x^2 - iy) dz$ along the paths (a) $y = x$ and (b) $y = x^2$. (U.P.T.U., 2010)
2. Evaluate $\int_{1-i}^{2+i} (2x + iy + 1) dz$, along the two paths:
(i) $x = t + 1, y = 2t^2 - 1$ (ii) the straight line joining $1 - i$ and $2 + i$. (U.P.T.U., 2010)
(U.P.T.U., 2006)
3. Evaluate $\int_{1-i}^{2+3i} (z^2 + z) dz$ along the line joining the points $(1, -1)$ and $(2, 3)$. (V.T.U., 2004)
4. Show that for every path between the limits, $\int_{-2}^{-2+i} (2+z)^2 dz = -i/3$. (Delhi, 2002)
5. Show that $\oint_C (z+1) dz = 0$, where C is the boundary of the square whose vertices are at the points $z = 0, z = 1, z = 1+i$ and $z = i$. (Rohtak, 2006)
6. Evaluate $\int_C |z| dz$, where C is the contour
(i) straight line from $z = -i$ to $z = i$.
(ii) left half of the unit circle $|z| = 1$ from $z = -i$ to $z = i$.
(iii) circle given by $|z + 1| = 1$ described in the clockwise sense.
7. Find the value of $\int_0^{1+i} (x - y + ix^2) dz$
(i) along the straight line from $z = 0$ to $z = 1 + i$
(ii) along real axis from $z = 0$ to $z = 1$ and then along a line parallel to the imaginary axis from $z = 1$ to $z = 1 + i$. (U.P.T.U., 2003)
8. Prove that $\int_C dz/z = -\pi i$ or πi , according as C is the semi-circular arc $|z| = 1$ above or below the real axis. (Rohtak, 2005)
9. Evaluate $\int_C (z - z^2) dz$, where C is the upper half of the circle $|z| = 1$.
What is the value of this integral if C is the lower half of the above circle?

20.13 CAUCHY'S THEOREM

If $f(z)$ is an analytic function and $f'(z)$ is continuous at each point within and on a closed curve C , then $\oint_C f(z) dz = 0$.

Writing $f(z) = u(x, y) + iv(x, y)$ and noting that $dz = dx + idy$

$$\oint_C f(z) dz = \oint_C (u dx - v dy) + i \oint_C (v dx + u dy) \quad \dots(1)$$

Since $f'(z)$ is continuous, therefore, $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ are also continuous in the region D enclosed by C .

Hence the Green's theorem (p. 376) can be applied to (1), giving

$$\oint_C f(z) dz = - \iint_D \left[\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right] dx dy + i \iint_D \left[\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right] dx dy \quad \dots(2)$$

Now $f(z)$ being analytic, u and v necessarily satisfy the Cauchy-Riemann equations and thus the integrands of the two double integrals in (2) vanish identically.

Hence
$$\oint_C f(z) dz = 0.$$

Obs. 1. The Cauchy-Riemann equations are precisely the conditions for the two real integrals in (1) to be independent of the path. Hence the line integral of a function $f(z)$ which is analytic in the region D , is independent of the path joining any two points of D .

Obs. 2. Extension of Cauchy's theorem. If $f(z)$ is analytic in the region D between two simple closed curves C and C_1 , then
$$\oint_C f(z) dz = \oint_{C_1} f(z) dz.$$

To prove this, we need to introduce the cross-cut AB . Then $\oint f(z)dz = 0$ where the path is as indicated by arrows in Fig. 20.17, i.e., along AB —along C_1 in clockwise sense and along BA —along C in anti-clockwise sense.

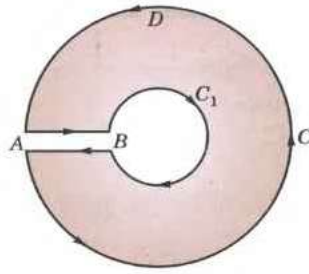


Fig. 20.17(a)

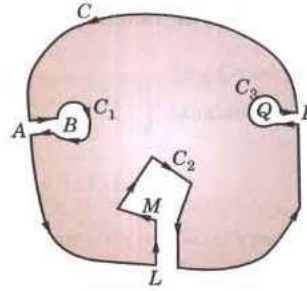


Fig. 20.17(b)

$$\text{i.e.,} \quad \int_{AB} f(z)dz + \int_{C_1} f(z)dz + \int_{BA} f(z)dz + \int_C f(z)dz = 0$$

But, since the integrals along AB and along BA cancel, it follows that

$$\int_C f(z)dz + \int_{C_1} f(z)dz = 0$$

Reversing the direction of the integral around C_1 and transposing, we get

$$\int_C f(z)dz + \int_{C_1} f(z)dz \text{ each integration being taken in the anti-clockwise sense.}$$

If $C_1, C_2, C_3, \dots$ be any number of closed curves within C (Fig. 20.17(b)), then

$$\oint_C f(z)dz = \oint_{C_1} f(z)dz + \oint_{C_2} f(z)dz + \oint_{C_3} f(z)dz + \dots$$

20.14 CAUCHY'S INTEGRAL FORMULA

If $f(z)$ is analytic within and on a closed curve and if a is any point within C , then

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)dz}{z-a}$$

Consider the function $f(z)/(z-a)$ which is analytic at all points within C except at $z=a$. With the point a as centre and radius r , draw a small circle C_1 lying entirely within C .

Now $f(z)/(z-a)$ being analytic in the region enclosed by C and C_1 , we have by Cauchy's theorem,

$$\begin{aligned} \oint_C \frac{f(z)}{z-a} dz &= \oint_{C_1} \frac{f(z)}{z-a} dz \\ &= \oint_{C_1} \frac{f(a + re^{i\theta})}{re^{i\theta}} \cdot ire^{i\theta} d\theta = i \oint_{C_1} f(a + re^{i\theta}) d\theta \end{aligned} \quad \left\{ \begin{array}{l} \text{For any point on } C_1, \\ z-a = re^{i\theta} \text{ and } dz = ire^{i\theta} d\theta \end{array} \right. \quad \dots(1)$$

In the limiting form, as the circle C_1 shrinks to the point a , i.e., as $r \rightarrow 0$, the integral (1) will approach to

$$\oint_C \frac{f(z)}{z-a} dz = 2\pi i f(a) \quad \text{Thus} \quad \oint_C \frac{f(z)}{z-a} dz = 2\pi i f(a)$$

$$\text{i.e.,} \quad f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a} dz \quad \dots(2)$$

which is the desired Cauchy's integral formula.

(V.T.U., 2011 S)

Cor. Differentiating both sides of (2) w.r.t. a ,

$$f'(a) = \frac{1}{2\pi i} \oint_C \frac{\partial}{\partial a} \left[\frac{f(z)}{z-a} \right] dz = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-a)^2} dz \quad \dots(3)$$

Similarly,

$$f''(a) = \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^3} dz \quad \dots(4)$$

and in general,

$$f^n(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^{n+1}} dz. \quad \dots(5)$$

Thus it follows from the results (2) to (5) that if a function $f(z)$ is known to be analytic on the simple closed curve C then the values of the function and all its derivatives can be found at any point of C . Incidentally, we have established a remarkable fact that **an analytic function possesses derivatives of all orders and these are themselves all analytic.**

Example 20.20. Evaluate $\int_C \frac{z^2 - z + 1}{z - 1} dz$, where C is the circle

$$(i) |z| = 1, \quad (ii) |z| = \frac{1}{2}. \quad (S.V.T.U., 2007)$$

Solution. (i) Here $f(z) = z^2 - z + 1$ and $a = 1$.

Since $f(z)$ is analytic within and on circle $C : |z| = 1$ and $a = 1$ lies on C .

$\therefore$ by Cauchy's integral formula $\frac{1}{2\pi i} \int_C \frac{f(z)}{z - a} dz = f(a) = 1$ i.e., $\int_C \frac{z^2 - z + 1}{z - 1} dz = 2\pi i$.

(ii) In this case, $a = 1$ lies outside the circle $C : |z| = 1/2$. So $(z^2 - z + 1)/(z - 1)$ is analytic everywhere within C .

$\therefore$ by Cauchy's theorem $\int_C \frac{z^2 - z + 1}{z - 1} dz = 0$.

Example 20.21. Evaluate, using Cauchy's integral formula:

$$(i) \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz \text{ where } C \text{ is the circle } |z| = 3 \quad (U.P.T.U., 2010)$$

$$(ii) \oint_C \frac{\cos \pi z}{z^2 - 1} dz \text{ around a rectangle with vertices } 2 \pm i, -2 \pm i$$

$$(iii) \oint_C \frac{e^{iz}}{z^2 + 1} dz \text{ where } C \text{ is the circle } |z| = 3. \quad (U.P.T.U., 2009)$$

Solution. (i) $f(z) = \sin \pi z^2 + \cos \pi z^2$ is analytic within the circle $|z| = 3$ and the two singular points $z = 1$ and $z = 2$ lie inside this circle.

$$\begin{aligned} \therefore \oint_C \frac{f(z)dz}{(z-1)(z-2)} &= \oint_C (\sin \pi z^2 + \cos \pi z^2) \left(\frac{1}{z-2} - \frac{1}{z-1} \right) \\ &= \oint_C \frac{(\sin \pi z^2 + \cos \pi z^2)}{z-2} dz - \oint_C \frac{(\sin \pi z^2 + \cos \pi z^2)}{z-1} dz \\ &= 2\pi i [\sin \pi(2)^2 + \cos \pi(2)^2] - 2\pi i [\sin \pi(1)^2 + \cos \pi(1)^2] \\ &= 2\pi i (0 + 1) - 2\pi i (0 - 1) = 4\pi i \end{aligned}$$

[By Cauchy's integral formula]

(ii) $f(z) = \cos \pi z$ is analytic in the region bounded by the given rectangle and the two singular points $z = 1$ and $z = -1$ lie inside this rectangle. (Fig. 20.18)

$$\begin{aligned} \therefore \oint_C \frac{\cos \pi z}{z^2 - 1} dz &= \frac{1}{2} \oint_C \left(\frac{1}{z-1} - \frac{1}{z+1} \right) \cos \pi z dz \\ &= \frac{1}{2} \oint_C \frac{\cos \pi z}{z-1} dz - \oint_C \frac{\cos \pi z}{z+1} dz \\ &= \frac{1}{2} [2\pi i \cos \pi(1)] - \frac{1}{2} [2\pi i \cos \pi(-1)] = 0. \end{aligned}$$

[By Cauchy's integral formula]

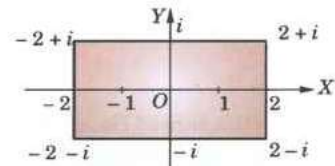


Fig. 20.18

(iii) $f(z) = e^{tz}$ is analytic within the circle $|z| = 3$.

The singular points are given by $z^2 + 1 = 0$ i.e., $z = i$ and $z = -i$ which lie within this circle.

$$\begin{aligned} \therefore \oint_C \frac{e^{tz}}{z^2 + 1} dz &= \oint_C \frac{1}{2i} \left(\frac{1}{z-i} - \frac{1}{z+i} \right) e^{tz} dz = \frac{1}{2i} \left\{ \oint_C \frac{e^{tz}}{z-i} dz - \oint_C \frac{e^{tz}}{z+i} dz \right\} \\ &= \frac{1}{2i} \{ 2\pi i e^{t(i)} - 2\pi i e^{t(-i)} \} \quad [\text{By Cauchy's integral formula}] \\ &= 2\pi i \left(\frac{e^{it} - e^{-it}}{2i} \right) = 2\pi i \sin t. \end{aligned}$$

Example 20.22. Evaluate

(i) $\oint_C \frac{\sin^2 z}{(z - \pi/6)^3} dz$, where C is the circle $|z| = 1$ (Rohtak, 2005)

(ii) $\oint_C \frac{e^{2z}}{(z+i)^4} dz$, where C is the circle $|z| = 3$ (U.P.T.U., 2008)

(iii) $\oint_C \frac{e^z}{(z^2 + \pi^2)^2} dz$, where C is $|z| = 4$. (U.P.T.U., 2008; J.N.T.U., 2000)

Solution. (i) $f(z) = \sin^2 z$ is analytic inside the circle $C: |z| = 1$ and the point $a = \pi/6$ (≈ 0.5 approx.) lies within C .

$$\therefore \text{by Cauchy's integral formula } f''(a) = \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^3} dz,$$

we get
$$\oint_C \frac{\sin^2 z}{(z - \pi/6)^3} dz = \pi i \left[\frac{d^2}{dz^2} (\sin^2 z) \right]_{z=\pi/6}$$

$$= \pi i (2 \cos 2z)_{z=\pi/6} = 2\pi i \cos \pi/3 = \pi i.$$

(ii) $f(z) = e^{2z}$ is analytic within the circle $C: |z| = 3$. Also $z = -1$ lies inside C .

$$\therefore \text{By Cauchy's integral formula: } f'''(a) = \frac{3!}{2\pi i} \int_C \frac{f(z) dz}{(z-a)^4}$$

we get
$$\int_C \frac{e^{2z}}{(z+1)^4} dz = \frac{2\pi i}{6} \left[\frac{d^3(e^{2z})}{dz^3} \right]_{z=-1} = \frac{\pi i}{3} [8e^{2z}]_{z=-1} = \frac{8\pi i}{3} e^{-2}$$

(iii) $\frac{e^z}{(z^2 + \pi^2)^2} = \frac{e^z}{(z + \pi i)^2 (z - \pi i)^2}$ is not analytic at $z = \pm \pi i$.

However both $z = \pm \pi i$ lie within the circle $|z| = 4$.

Now
$$\frac{1}{(z + \pi i)^2 (z - \pi i)^2} = \frac{A}{z + \pi i} + \frac{B}{(z + \pi i)^2} + \frac{C}{z - \pi i} + \frac{D}{(z - \pi i)^2}$$

where $A = 7/2\pi^3 i$, $C = -7/2\pi^3 i$, $B = D = -1/4\pi^2$

$$\begin{aligned} \therefore \int_C \frac{e^z}{(z^2 + \pi^2)^2} dz &= \frac{7}{2\pi^3 i} \left\{ \int_C \frac{e^z}{z + \pi i} dz - \int_C \frac{e^z}{z - \pi i} dz \right\} - \frac{1}{4\pi^2} \left\{ \int_C \frac{e^z}{(z + \pi i)^2} dz + \int_C \frac{e^z}{(z - \pi i)^2} dz \right\} \\ &= \frac{7}{2\pi^3 i} [2\pi i f(-\pi i) - 2\pi i f(\pi i)] - \frac{1}{4\pi^2} [2\pi i f'(-\pi i) + 2\pi i f'(\pi i)] \quad \text{where } f(z) = e^z. \\ &= \frac{7}{\pi^2} (e^{-\pi i} - e^{\pi i}) - \frac{i}{2\pi} (e^{-\pi i} + e^{\pi i}) = -\frac{14i}{\pi^2} \left(\frac{e^{\pi i} - e^{-\pi i}}{2i} \right) - \frac{i}{\pi} \left(\frac{e^{\pi i} + e^{-\pi i}}{2} \right) \quad [\S 19.9] \\ &= -\frac{14i}{\pi^2} \sin \pi - \frac{i}{\pi} \cos \pi = \frac{i}{\pi}. \end{aligned}$$

Example 20.23. Verify Cauchy's theorem by integrating e^{iz} along the boundary of the triangle with the vertices at the points $1 + i$, $-1 + i$ and $-1 - i$. (U.P.T.U., 2006)

Solution. The boundary of the given triangle consists of three lines AB , BC , CA . (Fig. 29.19).

$$\oint_{ABC} e^{iz} dz = \int_{AB} e^{iz} dz + \int_{BC} e^{iz} dz + \int_{CA} e^{iz} dz$$

Now $\int_{AB} e^{iz} dz = \int_1^{-1} e^{i(x+i)} dx \quad \left| \begin{array}{l} \because \text{Along } AB : y = 1 \\ \therefore z = x + i \text{ and } dz = dx \end{array} \right.$

$$= \int_1^{-1} e^{(ix-1)} dx = \left| \frac{e^{ix-1}}{i} \right|_1^{-1} = \frac{e^{-i-1} - e^{i-1}}{i}$$

$$\int_{BC} e^{iz} dz = \int_1^{-1} e^{i(-1+iy)} idy \quad \left| \begin{array}{l} \because \text{Along } BC : x = -1 \\ \therefore z = -1 + iy, dz = idy \end{array} \right.$$

$$= i \int_1^{-1} e^{-i-y} dy = i \left| \frac{e^{-i-y}}{-1} \right|_1^{-1} = \frac{e^{-i+1} - e^{-i-1}}{i}$$

$$\int_{CA} e^{iz} dz = \int_{-1}^1 e^{i(1+ix)} (1+i) dx$$

$$= (1+i) \frac{e^{(i-1)} - e^{-(i-1)}}{i(1+i)} = \frac{e^{i-1} - e^{-i+1}}{i}$$

$$\text{Thus from (i)} \quad \oint_{ABC} e^{iz} dz = \frac{e^{-i-1} - e^{i-1}}{i} + \frac{e^{-i+1} - e^{-i-1}}{i} + \frac{e^{i-1} - e^{-i+1}}{i} = 0 \quad \dots(ii)$$

Also since $f(z) = e^{iz}$ is analytic everywhere,

$$\therefore \text{ by Cauchy's theorem } \oint f(z) = 0 \quad \dots(iii)$$

Hence from (ii) and (iii), $\oint_{ABC}$ Cauchy's theorem is verified.

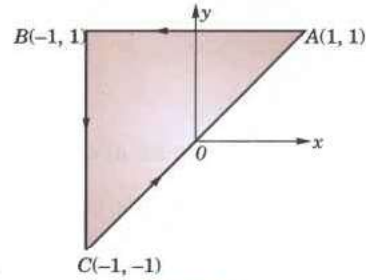


Fig. 20.19

Example 20.24. If $F(\zeta) = \oint_C \frac{4z^2 + z + 5}{z - \zeta} dz$, where C is the ellipse $(x/2)^2 + (y/3)^2 = 1$, find the value of (a) $F(3.5)$; (b) $F(i)$, $F''(-1)$ and $F'''(-i)$. (Bhopal, 2009; Marathwada, 2008; Mumbai, 2006)

Solution. (a) $F(3.5) = \oint_C \frac{z^3 + z + 1}{z^2 - 7z + 2} dz$

Since $\zeta = 3.5$ is the only singular point of $(4z^2 + z + 5)/(z - 3.5)$ and it lies outside the ellipse C , therefore, $(4z^2 + z + 5)/(z - 3.5)$ is analytic everywhere within C .

Hence by Cauchy's theorem,

$$\oint_C \frac{4z^2 + z + 5}{z - 3.5} dz = 0, \text{ i.e., } F(3.5) = 0.$$

(b) Since $f(z) = 4z^2 + z + 5$ is analytic within C and $\zeta = i, -1$ and $-i$ all lie within C , therefore, by Cauchy's integral formula

$$f(\zeta) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - \zeta} dz$$

$$\begin{aligned}
 \text{i.e.,} \quad & \oint_C \frac{4z^2 + z + 5}{z - \zeta} dz = 2\pi i(4\zeta^2 + \zeta + 5) \\
 \text{i.e.,} \quad & F(\zeta) = 2\pi i(4\zeta^2 + \zeta + 5) \\
 \therefore \quad & F'(\zeta) = 2\pi i(8\zeta + 1) \text{ and } F''(\zeta) = 16\pi i \\
 \text{Thus} \quad & F(i) = 2\pi i(-4 + i + 5) = 2\pi i(i - 1) \\
 & F'(-1) = 2\pi i[8(-1) + 1] = -14\pi i \text{ and } F''(-i) = 16\pi i.
 \end{aligned}$$

20.15 (1) CONVERSE OF CAUCHY'S THEOREM: MORERA'S THEOREM*

If $f(z)$ is continuous in a region D and $\oint_C f(z) dz = 0$ around every simple closed curve C in D , then $f(z)$ is analytic in D .

Since $\oint f(z) dz = 0$, then the line integral of $f(z)$ from a fixed point z_0 to a variable point z must be independent of the path and hence must be a function of z only. Thus

$$\int_{z_0}^z f(z) dz = \phi(z), \text{ (say),}$$

Let $\phi(z) = U + iV$ and $f(z) = u + iv$

$$\text{Then } U + iV = \int_{(x_0, y_0)}^{(x, y)} (u + iv)(dx + idy) = \int_{(x_0, y_0)}^{(x, y)} (udx - vdy) + i \int_{(x_0, y_0)}^{(x, y)} (vdx + udy)$$

$$\therefore U = \int_{(x_0, y_0)}^{(x, y)} (udx - vdy), V = \int_{(x_0, y_0)}^{(x, y)} (vdx + udy)$$

Differentiating under the integral sign,

$$\frac{\partial U}{\partial x} = u, \frac{\partial U}{\partial y} = -v, \frac{\partial V}{\partial x} = v, \frac{\partial V}{\partial y} = u \quad \therefore \frac{\partial U}{\partial x} = \frac{\partial V}{\partial y}, \frac{\partial U}{\partial y} = -\frac{\partial V}{\partial x}$$

Thus U and V satisfy C-R equations.

Also, since $f(z)$ is given to be continuous, u and v and therefore, $\partial U/\partial x$, $\partial U/\partial y$, $\partial V/\partial x$, $\partial V/\partial y$, are also continuous.

$\therefore \phi(z)$ is an analytic function and

$$\phi'(z) = \frac{\partial U}{\partial x} + i \frac{\partial V}{\partial x} = u + iv = f(z).$$

Thus, $f(z)$ is the derivative of an analytic function $\phi(z)$. Hence $f(z)$ is analytic by § 20.14 Cor.

(2) **Cauchy's inequality**[†]. If $f(z)$ is analytic within and on the circle $C: |z - a| = r$, then

$$|f^n(a)| \leq \frac{Mn!}{r^n} \quad \dots(I)$$

where M is the maximum value of $|f(z)|$ on C .

From (5) of § 20.14, we get

$$\begin{aligned}
 |f^n(a)| &= \frac{n!}{2\pi} \left| \oint_C \frac{f(z) dz}{(z - a)^{n+1}} \right| \\
 &\leq \frac{n!}{2\pi} \cdot \frac{M}{r^{n+1}} \oint_C |z| \quad [\because |f(z)| < M] \\
 &= \frac{n! M}{2\pi r^{n+1}} \oint_C ds = \frac{Mn!}{2\pi r^{n+1}} 2\pi r = \frac{Mn!}{r^n} \quad (U.P.T.U., 2005)
 \end{aligned}$$

(3) **Liouville's theorem**[§]. If $f(z)$ is analytic and bounded for all z in the entire complex plane, then $f(z)$ is a constant. (U.P.T.U., 2008)

* Named after the Italian mathematician, *Giacinto Morera* (1856–1909) who worked in Turin.

† See footnote p. 144

§ See footnote p. 573.

Taking $n = 1$ and replacing a by z , (1) gives

$$|f'(z)| \leq M/r$$

As $r \rightarrow \infty$, it gives $f'(z) = 0$ i.e., $f(z)$ is constant for all z .

(4) Poisson's integral formulae. If $f(z)$ is analytic within and on the circle $C: |z| = \rho$ and $z = re^{i\theta}$ is any point within C , then

$$f(re^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} \frac{\rho^2 - r^2}{\rho^2 - 2\rho r \cos(\theta - \phi) + r^2} f(\rho e^{i\phi}) d\phi$$

$$\text{By Cauchy's integral formula, } f(z) = \frac{1}{2\pi i} \oint_C \frac{f(w)}{w - z} dw \quad \dots(1)$$

As the inverse of the point z w.r.t. C lies outside C and is given by $\rho^2/\bar{z}$.

[See footnote p. 685]

$\therefore$ by Cauchy's theorem,

$$0 = \frac{1}{2\pi i} \int_C \frac{f(w)}{w - \rho^2/\bar{z}} dw \quad \dots(2)$$

$$\text{Subtracting (2) from (1), } f(z) = \frac{1}{2\pi i} \int_C \left(\frac{1}{w - z} - \frac{1}{w - \rho^2/\bar{z}} \right) f(w) dw$$

$$= \frac{1}{2\pi i} \oint_C \frac{z\bar{z} - \rho^2}{z\bar{w}^2 - (z\bar{z} + \rho^2)w + z\rho^2} f(w) dw \quad \dots(3)$$

Taking $w = \rho e^{i\phi}$ and noting that $\bar{z} = re^{-i\theta}$, (3) gives

$$\begin{aligned} f(re^{i\theta}) &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{(r^2 - \rho^2) f(\rho e^{i\phi}) \cdot \rho i e^{i\phi} d\phi}{re^{-i\theta} \cdot \rho^2 e^{2i\phi} - (r^2 + \rho^2) \rho e^{i\phi} + re^{-i\theta} \rho^2} \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{(\rho^2 - r^2) f(\rho e^{i\phi}) d\phi}{\rho^2 + r^2 - \rho r [e^{i(\theta - \phi)} + e^{-i(\theta - \phi)}]} = \frac{1}{2\pi} \int_0^{2\pi} \frac{(\rho^2 - r^2) f(\rho e^{i\phi}) d\phi}{\rho^2 - 2\rho r \cos(\theta - \phi) + r^2} \end{aligned} \quad \dots(4)$$

This is called *Poisson's integral formula** for a circle. It expresses the values of a harmonic function within a circle in terms of its values on the boundary.

Writing $f(re^{i\theta}) = u(r, \theta) + iv(r, \theta)$ and $f(\rho e^{i\phi}) = u(\rho, \phi) + iv(\rho, \phi)$ in (4) and equating real and imaginary parts from both sides, we get the formulae:

$$u(r, \theta) = \int_0^{2\pi} \frac{(\rho^2 - r^2) u(\rho, \phi) d\phi}{\rho^2 - 2\rho r \cos(\theta - \phi) + r^2} \quad \dots(5)$$

and

$$v(r, \theta) = \int_0^{2\pi} \frac{(\rho^2 - r^2) v(\rho, \phi) d\phi}{\rho^2 - 2\rho r \cos(\theta - \phi) + r^2} \quad \dots(6)$$

PROBLEMS 20.6

- Evaluate $\oint_C (z - a)^{-1} dz$, where C is a simple closed curve and the point $z = a$ is (i) outside C ,
(ii) inside C .
- Evaluate $\oint_C \frac{dz}{(z - a)^n}$, $n = 2, 3, 4, \dots$, where C is a closed curve containing the point $z = a$.
- Evaluate (i) $\oint_C \frac{e^z}{z^2 + 1} dz$, where C is the circle $|z| = 1/2$. (P.T.U., 2010)
(ii) $\oint_C \frac{e^{3iz}}{(z + \pi)^3} dz$, where C is the circle $|z - \pi| = 3$. (U.P.T.U., 2007)

* Named after the French mathematician *Simeon Denis Poisson* (1781–1842) who was a professor in Paris and made contributions to partial differential equations, potential theory and probability.

4. Use Cauchy's integral formula to calculate:

(i) $\oint_C \frac{3z-5}{z^2+2z} dz$, where C is $|z|=1$. (P.T.U., 2005 S) (ii) $\oint_C \frac{z^2+1}{z(2z+1)} dz$, where C is $|z|=1$.

(iii) $\oint_C \frac{\sin \pi z + \cos \pi z}{(z-1)(z-2)} dz$ where C is $|z|=4$. (U.P.T.U., 2008)

5. Evaluate (a) $\oint_C \frac{z^3-2z+1}{(z-i)^2} dz$ where C is $|z|=2$.

(b) $\oint_C \frac{e^{-z}}{(z-1)(z-2)^2} dz$ where C is $|z|=3$. (Rohtak, 2003)

6. Evaluate, using Cauchy's integral formulae:

(i) $\oint_C \frac{z}{z^2-3z+2} dz$, where C is $|z-2|=\frac{1}{2}$. (U.P.T.U., 2009; Hissar, 2007; Madras, 2000)

(ii) $\oint_C \frac{e^z dz}{(z+1)^2}$, where C is $|z-1|=3$. (Bhopal, 2009)

(iii) $\oint_C \frac{\log z}{(z-1)^3} dz$ where C is $|z-1|=\frac{1}{2}$. (J.N.T.U., 2003)

7. Evaluate $f'(2)$ and $f'(3)$ where $f(z) = \oint_C \frac{2z^2-z-2}{z-a} dz$ and C is the circle $|z|=2.5$.

8. If $\phi(\zeta) = \oint_C \frac{3z^2+7z+1}{z-\zeta} dz$, where C is the circle $|z|=2$ find the values of

(i) $\phi(3)$, (ii) $\phi'(1-i)$, (iii) $\phi''(1-i)$. (Mumbai, 2006)

9. Evaluate $\oint_C \frac{z^3+z+1}{z^2-7z+2} dz$, where C is the ellipse $4x^2+9y^2=1$. (Rohtak, 2006)

10. Verify Cauchy's theorem for the integral of z^3 taken over the boundary of the (i) rectangle with vertices $-1, 1, 1+i, -1+i$; (ii) triangle with vertices $(1, 2), (1, 4), (3, 2)$. (V.T.U., 2003)

20.16 (1) SERIES OF COMPLEX TERMS

Let $(a_1 + ib_1) + (a_2 + ib_2) + \dots + (a_n + ib_n) + \dots$... (1)

be an infinite series of complex terms; a 's and b 's being real numbers. If the series Σa_n and Σb_n converge to the sums A and B , then series (1) is said to **converge** to the sum $A + iB$. Also if (1) is a convergent series, then

$\lim_{n \rightarrow \infty} (a_n + ib_n) = 0$.

The series (1) is said to be **absolutely convergent** if the series

$|a_1 + ib_1| + |a_2 + ib_2| + \dots + |a_n + ib_n| + \dots$

is convergent. Since $|a_n|$ and $|b_n|$ are both $\leq |a_n + ib_n|$, it follows that an absolutely convergent series is convergent.

Next let the series of functions $u_1(z) + u_2(z) + \dots + u_n(z) + \dots$... (2)

converge to the sum $S(z)$ and $S_n(z)$ be the sum of its first n terms. Then the series (2) is said to be **uniformly convergent** in a region R , if corresponding to any positive number ϵ , there exists a positive number N , depending on ϵ , but not on z , such that for every z in R .

$|S(z) - S_n(z)| < \epsilon$ for $n > N$. [cf. Def. p. 389]

As in the case of real series (p. 390) **Weierstrass's M-test** holds for series of complex terms. So the series (2) is uniformly convergent in a region R if there is a convergent series of positive constants ΣM_n such that $|u_n(z)| \leq M_n$ for all z in R .

Also a uniformly convergent series of continuous complex functions is itself continuous and can be integrated term by term.

Obs. If a power series $\sum a_n z^n$ converges for $z = z_1$, then it converges absolutely for $|z| < |z_1|$.

Since $\sum a_n z_1^n$ converges, therefore, $\lim_{n \rightarrow \infty} a_n z_1^n = 0$ and so we can find a number k such that $|a_n z_1^n| < k$ for all n . Then

$$\sum a_n z^n = \sum |a_n z_1^n| \cdot |z/z_1|^n < \sum k t^n \text{ where } t = |z/z_1|.$$

But the series $\sum t^n$ converges for $t < 1$. Hence the series $\sum a_n z^n$ converges absolutely for $|z| < |z_1|$, i.e., if a circle with centre at the origin and radius $|z_1|$ be drawn, then the given series converges absolutely at all points inside the circle.

Such a circle $|z| = R$ within which series $\sum a_n z^n$ converges, is called the *circle of convergence* and R is called the *radius of convergence*.

A power series is uniformly convergent in any region which lies entirely within its circle of convergence.

(2) Taylor's series*. If $f(z)$ is analytic inside a circle C with centre at a , then for z inside C ,

$$f(z) = f(a) + f'(a)(z-a) + \frac{f''(a)}{2!}(z-a)^2 + \dots + \frac{f^n(a)}{n!}(z-a)^n + \dots \quad \dots(i)$$

Proof. Let z be any point inside C . Draw a circle C_1 with centre at a enclosing z (Fig. 20.20). Let t be a point on C_1 . We have

$$\begin{aligned} \frac{1}{t-z} &= \frac{1}{t-a-(z-a)} = \frac{1}{t-a} \left(1 - \frac{z-a}{t-a} \right)^{-1} \\ &= \frac{1}{t-a} \left[1 + \frac{z-a}{t-a} + \left(\frac{z-a}{t-a} \right)^2 + \dots + \left(\frac{z-a}{t-a} \right)^n + \dots \right] \quad \dots(ii) \end{aligned}$$

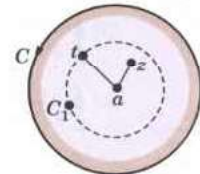


Fig. 20.20

As $|z-a| < |t-a|$, i.e., $|(z-a)/(t-a)| < 1$, this series converges uniformly. So, multiplying both sides of (ii) by $f(t)$, we can integrate over C_1 .

$$\therefore \oint_{C_1} \frac{f(t)}{t-z} dz = \oint_{C_1} \frac{f(t)}{t-a} dz + (z-a) \oint_{C_1} \frac{f(t)}{(t-a)^2} dt + \dots + (z-a)^n \oint_{C_1} \frac{f(t)}{(t-a)^{n+1}} dt + \dots \quad \dots(iii)$$

Since $f(t)$ is analytic on and inside C_1 , therefore, applying the formulae (2) to (5) of p. 697-698 (iii), we get (i) which is known as *Taylor's series*.

Obs. Another remarkable fact is that complex analytic functions can always be represented by power series of the form (i).

(3) Laurent's series†. If $f(z)$ is analytic in the ring-shaped region R bounded by two concentric circles C and C_1 of radii r and r_1 ($r > r_1$) and with centre at a , then for all z in R

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \dots + a_{-1}(z-a)^{-1} + a_{-2}(z-a)^{-2} + \dots$$

$$\text{where } a_n = \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(t)}{(t-a)^{n+1}} dt,$$

Γ being any curve in R , encircling C_1 (as in Fig. 20.21).

Proof. Introduce cross-cut AB , then $f(z)$ is analytic in the region D bounded by AB , C_1 described clockwise, BA and C described anti-clockwise (see Fig. 20.17). Then if z be any point in D , we have

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \left[\int_{AB} \frac{f(t)}{t-z} dt + \oint_{C_1} \frac{f(t)}{t-z} dt + \int_{BA} \frac{f(t)}{t-z} dt + \oint_C \frac{f(t)}{t-z} dt \right] \\ &= \frac{1}{2\pi i} \left[\oint_C \frac{f(t)}{t-z} dt - \oint_{C_1} \frac{f(t)}{t-z} dt \right] \quad \dots(i) \end{aligned}$$

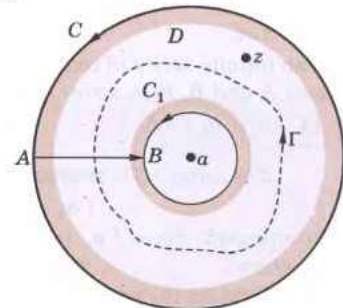


Fig. 20.21

where both C and C_1 are described anti-clockwise in (i) and integrals along AB and BA cancel (Fig. 20.21).

For the first integral in (i), expanding $1/(t-z)$ as in § 20.16 (2), we get

$$\frac{1}{2\pi i} \oint_C \frac{f(t)}{t-z} dt = \sum_{n=1}^{\infty} \frac{(z-a)^n}{2\pi i} \oint_C \frac{f(t)}{(t-a)^{n+1}} dt$$

* See footnote p. 145.

† Named after the French engineer and mathematician *Pierre Alphonse Laurent* (1813–1854) who published this theorem in 1843.

$$= \sum a_n (z-a)^n \text{ where } a_n = \frac{1}{2\pi i} \oint_C \frac{f(t)}{(t-a)^{n+1}} dt \quad \dots(ii)$$

For the second integral in (i), let t lie on C_1 . Then we write

$$\begin{aligned} \frac{1}{t-z} &= \frac{1}{(t-a)-(z-a)} = -\frac{1}{z-a} \left(1 - \frac{t-a}{z-a}\right)^{-1} \\ &= -\frac{1}{z-a} \left[1 + \frac{t-a}{z-a} + \left(\frac{t-a}{z-a}\right)^2 + \dots + \left(\frac{t-a}{z-a}\right)^{n-1} + \dots\right] \end{aligned}$$

As $|t-a| < |z-a|$, i.e., $|(t-a)/(z-a)| < 1$, this series converges uniformly. So multiplying both sides by $f(t)$ and integrating over C_1 , we get

$$-\frac{1}{2\pi i} \oint_C \frac{f(t)}{t-z} dt = \sum_{n=1}^{\infty} \frac{1}{(z-a)^n} \cdot \frac{1}{2\pi i} \oint_C (t-a)^{n-1} f(t) dt = \sum_{n=1}^{\infty} a_{-n} (z-a)^{-n} \quad \dots(iii)$$

where

$$a_{-n} = \frac{1}{2\pi i} \oint_{C_1} \frac{f(t)}{(t-a)^{-n+1}} dt$$

Substituting from (ii) and (iii) in (i), we get

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} a_{-n} (z-a)^{-n}. \quad \dots(iv)$$

Now $f(t)/(t-a)^{n+1}$ being analytic in the region between C and Γ , we can take the integral giving a_n over Γ . Similarly we can take the integral giving a_{-n} over Γ . Hence (iv) can be written as

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n \text{ where } a_n = \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(t)}{(t-a)^{n+1}} dt$$

which is known as *Laurent's series*.

Obs. 1. As $f(z)$ is not given to be analytic inside Γ , $a_n = \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(t)}{(t-a)^{n+1}} dt \neq \frac{f^n(a)}{n!}$

However, if $f(z)$ is analytic inside Γ , then $a_{-n} = 0$; $a_n = \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(t)}{(t-a)^{n+1}} dt = \frac{f^n(a)}{n!}$

and *Laurent's series reduces to Taylor's series*.

Obs. 2. To obtain Taylor's or Laurent's series, simply expand $f(z)$ by binomial theorem instead of finding a_n by complex integration which is quite complicated.

Obs. 3. Laurent series of a given analytic function $f(z)$ in its annulus of convergence is unique. There may be different Laurent series of $f(z)$ in two annuli with the same centre.

Example 20.25. Show that the series $z(1-z) + z^2(1-z) + z^3(1-z) + \dots \infty$ converges for $|z| < 1$. Determine whether it converges absolutely or not.

Solution. Let the sum of the first n terms of the series be s_n , so that

$$s_n = z - z^2 + z^2 - z^3 + z^3 - z^4 + \dots + z^n - z^{n+1} = z - z^{n+1}$$

For $|z| < 1$, $z^{n+1} \rightarrow 0$ as $n \rightarrow \infty$.

$\therefore \lim_{n \rightarrow \infty} (s_n) = z$, i.e., the given series converges for $|z| < 1$.

$$\begin{aligned} |s_n(z)| &= |z(1-z)| + |z^2(1-z)| + \dots + |z^n(1-z)| \\ &= |1-z| \{|z| + |z|^2 + |z|^3 + \dots + |z|^n\} \end{aligned}$$

$$\text{For } |z| < 1, \lim_{n \rightarrow \infty} |s_n(z)| = |1-z| \frac{|z|}{1-|z|}$$

[G.P.]

Hence the given series converges absolutely.

Example 20.26. Expand $\sin z$ in a Taylor's series about $z = 0$ and determine the region of convergence.

(P.T.U., 2009 S)

Solution. Given $f(z) = \sin z, f'(z) = \cos z, f''(z) = -\sin z, f'''(z) = -\cos z, \dots$
 $\therefore f(0) = 0, f'(0) = 1, f''(0) = 0, f'''(0) = 1$

By Taylor's series about $z = 0$, we have

$$f(z) = f(0) + \frac{(z-0)}{1!} f'(0) + \frac{(z-0)^2}{2!} f''(0) + \frac{(z-0)^3}{3!} f'''(0) + \dots$$

$$\text{i.e., } \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots + (-1)^{n-1} \frac{z^{2n-1}}{(2n-1)!} + \dots$$

$$\text{Hence } \sin z = \sum_{n=1}^{\infty} a_n (z-0)^{2n-1} \text{ where } a_n = \frac{(-1)^{n-1}}{(2n-1)!}$$

$$\text{Since } \rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(2n-1)!}{(2n+1)!} \right| = 0$$

Thus the radius of convergence of $f(z) = 1/\rho = \infty$

i.e., the region of convergence of $f(z)$ is all reals.

Example 20.27. Find Taylor's expansion of

$$(i) f(z) = \frac{1}{(z+1)^2} \text{ about the point } z = -i.$$

(V.T.U., 2009 S)

$$(ii) f(z) = \frac{2z^3 + 1}{z^2 + z} \text{ about the point } z = i.$$

(P.T.U., 2003)

Solution. (i) To expand $f(z)$ about $z = -i$, i.e., in powers of $z + i$, put $z + i = t$. Then

$$f(z) = \frac{1}{(t-i+1)^2} = (1-i)^{-2} [1 + t/(1-i)]^{-2} = \frac{i}{2} \left[1 - \frac{2t}{1-i} + \frac{3t^2}{(1-i)^2} - \frac{4t^3}{(1-i)^3} + \dots \right]$$

[Expanding by Binomial theorem]

$$= \frac{i}{2} \left[1 + \sum_{n=1}^{\infty} (-1)^n \frac{(n+1)(z+i)^n}{(1-i)^n} \right]$$

$$(ii) f(z) = \frac{2z^3 + 1}{z(z+1)} = 2z - 2 + \frac{2z+1}{z(z+1)} = (2i-2) + 2(z-i) + \frac{1}{z} + \frac{1}{z+1} \quad \dots(i)$$

[By partial fractions]

To expand $1/z$ and $1/(z+1)$ about $z = i$, put $z - i = t$, so that

$$\frac{1}{z} = \frac{1}{(t+i)} = \frac{1}{i} \left(1 + \frac{t}{i} \right)^{-1} \quad \text{[Expanding by Binomial theorem]}$$

$$= \frac{1}{i} \left[1 - \frac{t}{i} + \frac{t^2}{i^2} - \frac{t^3}{i^3} + \frac{t^4}{i^4} - \dots \infty \right] = \frac{1}{i} + \frac{t}{1} + \frac{t^2}{i^3} - \frac{t^3}{i^4} + \frac{t^4}{i^5} - \dots \infty$$

$$= -i + (z-i) + \sum_{n=2}^{\infty} (-1)^n \frac{(z-i)^n}{i^{n+1}} \quad \dots(ii)$$

$$\text{and } \frac{1}{z+1} = \frac{1}{t+i+1} = \frac{1}{1+i} \left(1 + \frac{t}{1+i} \right)^{-1} \quad \text{[Expanding by Binomial theorem]}$$

$$= \frac{1}{1+i} \left[1 - \frac{t}{1+i} + \frac{t^2}{(1+i)^2} - \frac{t^3}{(1+i)^3} + \frac{t^4}{(1+i)^4} - \dots \infty \right]$$

$$= \frac{1-i}{2} - \frac{t}{2i} + \left[\frac{t^2}{(1+i)^3} - \frac{t^3}{(1+i)^4} + \frac{t^4}{(1+i)^5} - \dots \infty \right] = \frac{1}{2} - \frac{i}{2} - \frac{z-i}{2i} + \sum_{n=2}^{\infty} (-1)^n \frac{(z-i)^n}{(1+i)^{n+1}} \quad \dots(iii)$$

Substituting from (ii) and (iii) in (i), we get

$$\begin{aligned} f(z) &= \left(2i - 2 - i + \frac{1}{2} - \frac{i}{2}\right) + \left(2 + 1 - \frac{1}{2i}\right)(z - i) + \sum_{n=2}^{\infty} (-1)^n \left\{ \frac{1}{i^{n+1}} + \frac{1}{(1+i)^{n+1}} \right\} (z - i)^n \\ &= \left(\frac{i}{2} - \frac{3}{2}\right) + \left(3 + \frac{i}{2}\right)(z - i) + \sum_{n=2}^{\infty} (-1)^n \left\{ \frac{1}{i^{n+1}} + \frac{1}{(1+i)^{n+1}} \right\} (z - i)^n. \end{aligned}$$

Example 20.28. Expand $f(z) = 1/[(z-1)(z-2)]$ in the region:

(a) $|z| < 1$, (b) $1 < |z| < 2$, (c) $|z| > 2$, (d) $0 < |z-1| < 1$.

(U.P.T.U., 2010 ; V.T.U., 2010 ; Bhopal, 2009)

Solution. (a) By partial fractions $\frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1}$... (i)

$$= -\frac{1}{2} \left(1 - \frac{z}{2}\right)^{-1} + (1-z)^{-1} \quad \dots (ii)$$

For $|z| < 1$, both $|z/2|$ and $|z|$ are less than 1. Hence (ii) gives on expansion

$$\begin{aligned} f(z) &= -\frac{1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots\right) + (1 + z + z^2 + z^3 + \dots) \\ &= \frac{1}{2} + \frac{3}{4}z + \frac{7}{8}z^2 + \frac{15}{16}z^3 + \dots \text{ which is a Taylor's series.} \end{aligned}$$

(b) For $1 < |z| < 2$, we write (i) as

$$f(z) = -\frac{1}{2} \frac{1}{(1-z/2)} - \frac{1}{z(1-z^{-1})} \quad \dots (iii)$$

and notice that both $|z/2|$ and $|z^{-1}|$ are less than 1. Hence (iii) gives on expansion

$$\begin{aligned} f(z) &= -\frac{1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots\right) - \frac{1}{z} (1 + z^{-1} + z^{-2} + z^{-3} + \dots) \\ &= \dots - z^{-4} - z^{-3} - z^{-2} - z^{-1} - \frac{1}{2} - \frac{1}{4}z - \frac{1}{8}z^2 - \frac{1}{16}z^3 - \dots \end{aligned}$$

which is a Laurent's series.

(c) For $|z| > 2$, we write (i) as

$$\begin{aligned} f(z) &= \frac{1}{z(1-2z^{-1})} - \frac{1}{z(1-z^{-1})} \\ &= z^{-1}(1 + 2z^{-1} + 4z^{-2} + 8z^{-3} + \dots) - z^{-1}(1 + z^{-1} + z^{-2} + z^{-3} + \dots) \\ &= \dots + 7z^{-4} + 3z^{-3} + z^{-2} + \dots \end{aligned}$$

(d) For $0 < |z-1| < 1$, we write (i) as

$$\begin{aligned} f(z) &= \frac{1}{(z-1)-1} - \frac{1}{z-1} \\ &= -(z-1)^{-1} - [1 - (z-1)]^{-1} \\ &= -(z-1)^{-1} - [1 + (z-1) + (z-1)^2 + (z-1)^3 + \dots]. \end{aligned}$$

Example 20.29. Find the Laurents' expansion of $f(z) = \frac{7z-2}{(z+1)z(z-2)}$ in the region $1 < z+1 < 3$.

(S.V.T.U., 2009 ; Anna, 2003 ; V.T.U., 2003)

Solution. Writing $z+1 = u$, we have

$$\begin{aligned} f(z) &= \frac{7(u-1)-2}{u(u-1)(u-1-2)} = \frac{7u-9}{u(u-1)(u-3)} \\ &= -\frac{3}{u} + \frac{1}{u-1} + \frac{2}{u-3} \quad (\text{splitting into partial fraction}) \\ &= -\frac{3}{u} + \frac{1}{u(1-1/u)} - \frac{2}{3(1-u/3)} = -\frac{3}{u} + \frac{1}{u} \left(1 - \frac{1}{u}\right)^{-1} - \frac{2}{3} \left(1 - \frac{u}{3}\right)^{-1} \end{aligned}$$

Since $1 < u < 3$ or $1/u < 1$ and $u/3 < 1$, expanding by Binomial theorem,

$$\begin{aligned} f(z) &= \frac{-3}{u} + \frac{1}{u} \left(1 + \frac{1}{u} + \frac{1}{u^2} + \frac{1}{u^3} + \dots \infty \right) - \frac{2}{3} \left(1 + \frac{u}{3} + \frac{u^2}{3^2} + \frac{u^3}{3^3} + \dots \infty \right) \\ &= -\frac{2}{u} + \frac{1}{u^2} + \frac{1}{u^3} + \dots \infty - \frac{2}{3} \left(1 + \frac{u}{3} + \frac{u^2}{3^2} + \frac{u^3}{3^3} + \dots \infty \right) \end{aligned}$$

$$\text{Hence } f(z) = -\frac{2}{z+1} + \frac{1}{(z+1)^2} + \frac{1}{(z+1)^3} + \dots \infty - \frac{2}{3} \left[1 + \frac{z+1}{3} + \frac{(z+1)^2}{3^2} + \frac{(z+1)^3}{3^3} + \dots \infty \right]$$

which is valid in the region $1 < z+1 < 3$.

20.17 (1) ZEROS OF AN ANALYTIC FUNCTION

Def. A zero of an analytic function $f(z)$ is that value of z for which $f(z) = 0$.

If $f(z)$ is analytic in the neighbourhood of a point $z = a$, then by Taylor's theorem

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \dots + a_n(z-a)^n + \dots \quad \text{where } a_n = \frac{f^{(n)}(a)}{n!}.$$

If $a_0 = a_1 = a_2 = \dots = a_{m-1} = 0$ but $a_m \neq 0$, then $f(z)$ is said to have a zero of order m at $z = a$.

When $m = 1$, the zero is said to be simple. In the neighbourhood of zero ($z = a$) of order m ,

$$\begin{aligned} f(z) &= a_m(z-a)^m + a_{m+1}(z-a)^{m+1} + \dots \infty \\ &= (z-a)^m \phi(z) \text{ where } \phi(z) = a_m + a_{m+1}(z-a) + \dots \end{aligned}$$

Then $\phi(z)$ is analytic and non-zero in the neighbourhood of $z = a$.

(2) Singularities of an analytic function

We have already defined a singular point of a function as the point at which the function ceases to be analytic.

(i) **Isolated singularity.** If $z = a$ is a singularity of $f(z)$ such that $f(z)$ is analytic at each point in its neighbourhood (i.e., there exists a circle with centre a which has no other singularity), then $z = a$ is called an isolated singularity.

In such a case, $f(z)$ can be expanded in a Laurent's series around $z = a$, giving

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \dots + a_{-1}(z-a)^{-1} + a_{-2}(z-a)^{-2} + \dots \quad \dots(1)$$

For example, $f(z) = \cot(\pi/z)$ is not analytic where $\tan(\pi/z) = 0$ i.e. at the points $\pi/z = 4\pi$ or $z = 1/n$ ($n = 1, 2, 3, \dots$).

Thus $z = 1, 1/2, 1/3, \dots$ are all isolated singularities as there is no other singularity in their neighbourhood.

But when n is large, $z = 0$ is such a singularity that there are infinite number of other singularities in its neighbourhood. Thus $z = 0$ is the non-isolated singularity of $f(z)$.

(ii) **Removable singularity.** If all the negative powers of $(z-a)$ in (1) are zero, then $f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n$.

Here the singularity can be removed by defining $f(z)$ at $z = a$ in such a way that it becomes analytic at $z = a$. Such a singularity is called a removable singularity.

Thus if $\lim_{z \rightarrow a} f(z)$ exists finitely, then $z = a$ is a removable singularity.

(iii) **Poles.** If all the negative powers of $(z-a)$ in (i) after the n th are missing, then the singularity at $z = a$ is called a pole of order n .

A pole of first order is called a simple pole.

(iv) **Essential singularity.** If the number of negative powers of $(z-a)$ in (1) is infinite, then $z = a$ is called an essential singularity. In this case, $\lim_{z \rightarrow a} f(z)$ does not exist.

Example 20.30. Find the nature and location of singularities of the following functions:

(i) $\frac{z - \sin z}{z^2}$

(ii) $(z+1) \sin \frac{1}{z-2}$

(iii) $\frac{1}{\cos z - \sin z}$

Solution. (i) Here $z = 0$ is a singularity.

$$\text{Also } \frac{z - \sin z}{z^2} = \frac{1}{z^2} \left\{ z - \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots \right) \right\} = \frac{z}{3!} - \frac{z^3}{5!} + \frac{z^5}{7!} - \dots$$

Since there are no negative powers of z in the expansion, $z = 0$ is a removable singularity.

$$\begin{aligned} \text{(ii) } (z+1) \sin \frac{1}{z-2} &= (t+2+1) \sin \frac{1}{t} \quad \text{where } t = z-2 \\ &= (t+3) \left\{ \frac{1}{t} - \frac{1}{3!t^3} + \frac{1}{5!t^5} - \dots \right\} = \left(1 - \frac{1}{3!t^2} + \frac{1}{5!t^4} - \dots \right) + \left(\frac{3}{t} - \frac{1}{2t^3} + \frac{3}{5!t^5} - \dots \right) \\ &= 1 + \frac{3}{t} - \frac{1}{6t^2} - \frac{1}{2t^3} + \frac{1}{120t^4} - \dots = 1 + \frac{3}{z-2} - \frac{1}{6(z-2)^2} - \frac{1}{2(z-2)^3} + \dots \end{aligned}$$

Since there are infinite number of terms in the negative powers of $(z-2)$, $z = 2$ is an essential singularity.

(iii) Poles of $f(z) = \frac{1}{\cos z - \sin z}$ are given by equating the denominator to zero, i.e., by $\cos z - \sin z = 0$ or $\tan z = 1$ or $z = \pi/4$. Clearly $z = \pi/4$ is a simple pole of $f(z)$.

Example 20.31. What type of singularity have the following functions :

$$(i) \frac{1}{1-e^z}$$

$$(ii) \frac{e^{2z}}{(z-1)^4}$$

$$(iii) \frac{e^{1/z}}{z^2}$$

(U.P.T.U., 2009)

Solution. (i) Poles of $f(z) = 1/(1-e^z)$ are found by equating to zero $1-e^z = 0$ or $e^z = 1 = e^{2n\pi i}$

$\therefore z = 2n\pi i$ ($n = 0, \pm 1, \pm 2, \dots$)

Clearly $f(z)$ has a simple pole at $z = 2\pi i$.

$$\begin{aligned} \text{(ii) } \frac{e^{2z}}{(z-1)^4} &= \frac{e^{2(t+1)}}{t^4} = \frac{e^2}{t^4} \cdot e^{2t} \quad \text{where } t = z-1 \\ &= \frac{e^2}{t^4} \left\{ 1 + \frac{2t}{1!} + \frac{(2t)^2}{2!} + \frac{(2t)^3}{3!} + \frac{(2t)^4}{4!} + \frac{(2t)^5}{5!} + \dots \right\} = e^2 \left\{ \frac{1}{t^4} + \frac{2}{t^3} + \frac{2}{t^2} + \frac{4}{3t} + \frac{2}{3} + \frac{4t}{15} + \dots \right\} \\ &= e^2 \left\{ \frac{1}{(z-1)^4} + \frac{2}{(z-1)^3} + \frac{2}{(z-1)^2} + \frac{4}{3(z-1)} + \frac{2}{3} + \frac{4}{15}(z-1) + \dots \right\} \end{aligned}$$

Since there are finite (4) number of terms containing negative powers of $(z-1)$,

$\therefore z = 1$ is a pole of 4th order.

$$\text{(iii) } f(z) = \frac{e^{1/z}}{z^2} = \frac{1}{z^2} \left\{ 1 + \frac{1}{1!z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots \right\} = z^{-2} + z^{-3} + \frac{z^{-4}}{2} + \dots \infty$$

Since there are infinite number of terms in the negative powers of z , therefore $f(z)$ has an essential singularity at $z = 0$.

PROBLEMS 20.7

- Obtain the expansion of $(z-1)/z^2$ in a Taylor's series in powers of $(z-1)$ and determine the region of convergence.
- Find the first three terms of the Taylor's series expansion of $f(z) = 1/(z^2+4)$ about $z = -i$. Also find the region of convergence. (U.P.T.U., 2006)
- Expand in Taylor's series (i) $(z-1)/(z+1)$ about the point $z = 1$. (Andhra, 2000)

(ii) $\cos z$ about the point $z = \pi/2$. (Marathwada, 2008) (iii) $\frac{1}{z^2-z-6}$ about (a) $z = -1$ (b) $z = 1$ (P.T.U., 2009)

- Expand the following functions in Laurent's series :

$$(i) f(z) = \frac{1}{z-z^2} \text{ for } 1 < |z+1| < 2.$$

(Madras, 2006)

$$(ii) f(z) = \frac{1}{(z-1)(z+3)} \text{ for } 1 < |z| < 3. \quad (J.N.T.U., 2006)$$

$$(iii) f(z) = z/[(z-1)(z-3)] \text{ for } |z-1| < 2. \quad (V.T.U., 2007)$$

$$5. \text{ Find the Laurent's expansion of (i) } \frac{e^z}{(z-1)^2}, \text{ about } z=1. \quad (Rohtak, 2006)$$

$$(ii) e^{2z}/(z-1)^3 \text{ about the singularity } z=1.$$

6. Expand the following functions in Laurent series.

$$(i) (z-1)/z^2 \text{ for } |z-1| > 1$$

$$(ii) \frac{1-\cos z}{z^3}, \text{ about } z=0. \quad (Rohtak, 2004)$$

7. Find the Laurent's series expansion of

$$(i) \frac{z^2-1}{z^2+5z+6} \text{ about } z=0 \text{ in the region } 2 < |z| < 3 \quad (V.T.U., 2011 S; Osmania, 2003)$$

$$(ii) \frac{z^2-6z-1}{(z-1)(z-3)(z+2)} \text{ in the region } 3 < |z+2| < 5$$

$$(iii) \frac{7z^2-9z-18}{z^3-9z} \text{ in the region } (a) |z| > 3 \quad (b) 0 < |z-3| < 3. \quad (V.T.U., 2010 S)$$

8. Find the Laurent's expansion of $1/[(z^2+1)(z^2+2)]$ for (a) $0 < |z| < 1$; (b) $1 < |z| < \sqrt{2}$; (c) $|z| > 2$.

Find the nature and location of the singularities of the following functions: (P.T.U., 2005)

$$9. \frac{1}{z(2-z)},$$

$$10. \sin(1/z). \quad (U.P.T.U., 2009)$$

$$11. \tan\left(\frac{1}{z}\right). \quad (P.T.U., 2006)$$

$$12. \frac{z^2-1}{(z-1)^3}. \quad (Osmania, 2003)$$

$$13. \frac{e^z}{(z-1)^4}.$$

$$14. \frac{\cot \pi z}{(z-a)^2}. \quad (U.P.T.U., 2008)$$

20.18 (1) RESIDUES

The coefficient of $(z-a)^{-1}$ in the expansion of $f(z)$ around an isolated singularity is called the **residue** of $f(z)$ at that point. Thus in the Laurent's series expansion of $f(z)$ around $z=a$ i.e., $f(z) = a_0(z-a) + a_1(z-a)^2 + \dots + a_{-1}(z-a)^{-1} + a_{-2}(z-a)^{-2} + \dots$, the residue of $f(z)$ at $z=a$ is a_{-1} .

$$\therefore \text{Res } f(a) = \frac{1}{2\pi i} \oint_C f(z) dz$$

$$\text{i.e.,} \quad \oint_C f(z) dz = 2\pi i \text{Res } f(a). \quad \dots(1)$$

(2) Residue Theorem

If $f(z)$ is analytic in a closed curve C except at a finite number of singular points within C , then $\oint_C f(z) dz = 2\pi i \times (\text{sum of the residues at the singular points within } C)$.

Let us surround each of the singular points $a_1, a_2, \dots, a_n$ by a small circle such that it encloses no other singular point (Fig. 20.22). Then these circles $C_1, C_2, \dots, C_n$ together with C , form a multiply connected region in which $f(z)$ is analytic.

$\therefore$ applying Cauchy's theorem, we have

$$\begin{aligned} \oint_C f(z) dz &= \oint_{C_1} f(z) dz + \oint_{C_2} f(z) dz + \dots + \oint_{C_n} f(z) dz \\ &= 2\pi i [\text{Res } f(a_1) + \text{Res } f(a_2) + \dots + \text{Res } f(a_n)] \text{ which is the desired result.} \end{aligned} \quad [\text{by } (1)]$$

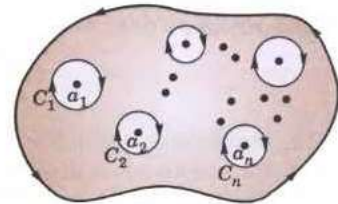


Fig. 20.22

20.19 CALCULATION OF RESIDUES

(1) If $f(z)$ has a simple pole at $z = a$, then

$$\text{Res } f(a) = \lim_{z \rightarrow a} [(z - a)f(z)]. \quad \dots(1)$$

Laurent's series in this case is

$$f(z) = c_0 + c_1(z - a) + c_2(z - a)^2 + \dots + c_{-1}(z - a)^{-1}$$

Multiplying throughout by $z - a$, we have

$$(z - a)f(z) = c_0(z - a) + c_1(z - a)^2 + \dots + c_{-1}.$$

Taking limits as $z \rightarrow a$, we get

$$\lim_{z \rightarrow a} [(z - a)f(z)] = c_{-1} = \text{Res } f(a).$$

(2) Another formula for $\text{Res } f(a)$:

Let $f(z) = \phi(z)/\psi(z)$, where $\psi(z) = (z - a)F(z)$, $F(a) \neq 0$.

Then

$$\begin{aligned} \lim_{z \rightarrow a} [(z - a)\phi(z)/\psi(z)] &= \lim_{z \rightarrow a} \frac{(z - a)[\phi(a) + (z - a)\phi'(a) + \dots]}{\psi(a) + (z - a)\psi'(a) + \dots} \\ &= \lim_{z \rightarrow a} \frac{\phi(a) + (z - a)\phi'(a) + \dots}{\psi'(a) + (z - a)\psi''(a) + \dots}, \quad \text{since } \psi(a) = 0 \end{aligned}$$

Thus

$$\text{Res } f(a) = \frac{\phi(a)}{\psi'(a)}.$$

(3) If $f(z)$ has a pole of order n at $z = a$, then

$$\text{Res } f(a) = \frac{1}{(n-1)!} \left\{ \frac{d^{n-1}}{dz^{n-1}} [(z - a)^n f(z)] \right\}_{z=a}$$

Here

$$f(z) = c_0 + c_1(z - a) + c_2(z - a)^2 + \dots + c_{-1}(z - a)^{-1} + \dots + c_{-n}(z - a)^{-n}.$$

Multiplying throughout by $(z - a)^n$, we get

$$(z - a)^n f(z) = c_0(z - a)^n + c_1(z - a)^{n+1} + c_2(z - a)^{n+2} + \dots + c_{-1}(z - a)^{n-1} + c_{-2}(z - a)^{n-2} + \dots + c_{-n}.$$

Differentiating both sides w.r.t. z , $n - 1$ times and putting $z = a$, we get

$$\left\{ \frac{d^{n-1}}{dz^{n-1}} [(z - a)^n f(z)] \right\}_{z=a} = (n-1)! c_{-1} \text{ whence follows the result.}$$

Obs. In many cases, the residue of a pole ($z = a$) can be found, by putting $z = a + t$ in $f(z)$ and expanding it in powers of t where $|t|$ is quite small.

Example 20.32. Find the sum of the residues of $f(z) = \frac{\sin z}{z \cos z}$ at its poles inside the circle $|z| = 2$.

(Rohtak, 2004)

Solution. $f(z)$ has simple poles at $z = 0, \pm \pi/2, \pm 3\pi/2, \dots$

Only the poles $z = 0$ and $z = \pm \pi/2$ lies inside $|z| = 2$.

$$\therefore \text{Res } f(0) = \lim_{z \rightarrow 0} [z \cdot f(z)] = \lim_{z \rightarrow 0} \left(\frac{\sin z}{\cos z} \right) = 0.$$

$$\begin{aligned} \text{Res } f(\pi/2) &= \lim_{z \rightarrow \pi/2} \left[\left(z - \frac{\pi}{2} \right) f(z) \right] = \lim_{z \rightarrow \pi/2} \left\{ \frac{(z - \pi/2) \sin z}{z \cos z} \right\} \\ &= \lim_{z \rightarrow \pi/2} \frac{(z - \pi/2) \cos z + \sin z}{\cos z - z \sin z} \\ &= \frac{1}{-\pi/2} = -\frac{2}{\pi} \end{aligned}$$

[Being $\frac{0}{0}$ form]

and $\text{Res } f(-\pi/2) = \lim_{z \rightarrow -\pi/2} \left\{ \frac{(z + \pi/2) \sin z}{z \cos z} \right\} = \lim_{z \rightarrow -\pi/2} \frac{(z + \pi/2) \cos z + \sin z}{\cos z - z \sin z} = \frac{-1}{-\pi/2} = \frac{2}{\pi}$

Hence sum of residues $= 0 - \frac{2}{\pi} + \frac{2}{\pi} = 0$.

Example 20.33. Determine the poles of the function

$$f(z) = z^2/(z-1)^2(z+2) \text{ and the residue at each pole. (S.V.T.U., 2008 ; J.N.T.U., 2005)}$$

Hence evaluate $\oint_C f(z) dz$, where C is the circle $|z| = 2.5$.

Solution. Since $\lim_{z \rightarrow -2} \{(z+2)f(z)\} = \lim_{z \rightarrow -2} \frac{z^2}{(z-1)^2} = \frac{4}{9}$.

which is finite and non-zero, the function has a simple pole at $z = -2$ and $\text{Res } f(-2) = 4/9$.

Also since $\lim_{z \rightarrow 1} \{(z-1)^2 f(z)\}$ is finite and non-zero, $f(z)$ has a pole of order two at $z = 1$.

$$\therefore \text{Res } f(1) = \frac{1}{1!} \left[\frac{d}{dz} \{(z-1)^2 f(z)\} \right]_{z=1} = \left[\frac{d}{dz} \left(\frac{z^2}{z+2} \right) \right]_{z=1} = \left[\frac{z^2 + 4z}{(z+2)^2} \right]_{z=1} = \frac{5}{9}.$$

[Otherwise writing $z = 1 + t$,

$$\begin{aligned} f(z) &= \frac{(1+t)^2}{t^2(3+t)} = \frac{1}{3t^2} (1+t)^2 (1+t/3)^{-1} = \frac{1}{3t^2} (1+t)^2 \left(1 - \frac{t}{3} + \frac{t^2}{9} - \dots \right) \\ &= \frac{1}{3t^2} \left(1 + \frac{5}{3}t + \frac{4}{9}t^2 - \dots \right) = \frac{1}{3t^2} + \frac{5}{9t} + \frac{4}{27} - \dots \end{aligned} \quad \dots(i)$$

$$\therefore \text{Res } f(1) = \text{coefficient of } \frac{1}{t} \text{ in (i)} = \frac{5}{9}.$$

Clearly $f(z)$ is analytic on $|z| = 2.5$ and at all points inside except the poles $z = -2$ and $z = 1$. Hence by residue theorem

$$\oint_C f(z) dz = 2\pi i [\text{Res } f(-2) + \text{Res } f(1)] = 2\pi i \left[\frac{4}{9} + \frac{5}{9} \right] = 2\pi i.$$

Example 20.34. Find the residue of $f(z) = \frac{z^3}{(z-1)^4(z-2)(z-3)}$ at its poles and hence evaluate $\oint_C f(z) dz$ where C is the circle $|z| = 2.5$. (U.P.T.U., 2003)

Solution. The poles of $f(z)$ are given by $(z-1)^4(z-2)(z-3) = 0$.

$\therefore z = 1$ is a pole of order 4, while $z = 2$ and $z = 3$ are simple poles.

$$\text{Res } f(1) = \frac{1}{3!} \frac{d^3}{dz^3} \left\{ (z-1)^4 \cdot \frac{z^3}{(z-1)^4(z-2)(z-3)} \right\}_{z=1} = \frac{1}{6} \frac{d^3}{dz^3} \left\{ \frac{z^3}{(z-2)(z-3)} \right\}_{z=1}$$

$\therefore z = 1$ is a pole of order 4, while $z = 2$ and $z = 3$ are simple poles.

$$\begin{aligned} \text{Res } f(1) &= \frac{1}{3!} \frac{d^3}{dz^3} \left\{ (z-1)^4 \cdot \frac{z^3}{(z-1)^4(z-2)(z-3)} \right\}_{z=1} = \frac{1}{6} \frac{d^3}{dz^3} \left\{ \frac{z^3}{(z-2)(z-3)} \right\}_{z=1} \\ &= \frac{1}{6} \frac{d^3}{dz^3} \left[z + 5 - \frac{8}{z-2} + \frac{27}{z-3} \right] = \frac{1}{6} \left[-8 \cdot \frac{(-1)^3 3!}{(z-2)^4} + \frac{27 \cdot (-1)^3 3!}{(z-2)^4} \right]_{z=1} \\ &= - \left[-8 + \frac{27}{16} \right] = \frac{101}{16}. \end{aligned}$$

$$\operatorname{Res} f(2) = \lim_{z \rightarrow 2} \left\{ (z-2) \cdot \frac{z^3}{(z-1)^4(z-2)(z-3)} \right\} = \lim_{z \rightarrow 2} \left\{ \frac{z^3}{(z-1)^4(z-3)} \right\} = \frac{8}{(1)^4(-1)} = -8$$

$$\operatorname{Res} f(3) = \lim_{z \rightarrow 3} \left\{ (z-3) \cdot \frac{z^3}{(z-1)^4(z-2)(z-3)} \right\} = \frac{27}{(2)^4 \cdot 1} = \frac{27}{16}$$

Now $\oint_C f(z) dz = 2\pi i [\operatorname{Res} f(1) + \operatorname{Res} f(2)]$ [$\because$ Pole $z = 3$ is outside C]

$$= 2\pi i \left(\frac{101}{16} - 8 \right) = \frac{-27\pi i}{8}.$$

Example 20.35. Evaluate

$$\oint_C \frac{z-3}{z^2+2z+5} dz, \text{ where } C \text{ is the circle}$$

(i) $|z| = 1$, (ii) $|z+1-i| = 2$, (iii) $|z+1+i| = 2$. (J.N.T.U., 2003)

Solution. The poles of $f(z) = \frac{z-3}{z^2+2z+5}$ are given by $z^2+2z+5=0$

i.e., by $z = \frac{-2 \pm \sqrt{4-20}}{2} = -1 \pm 2i$.

(i) Both the poles $z = -1+2i$ and $z = -1-2i$ lie outside the circle $|z| = 1$. Therefore, $f(z)$ is analytic everywhere within C .

Hence by Cauchy's theorem, $\oint_C \frac{z-3}{z^2+2z+5} dz = 0$.

(ii) Here only one pole $z = -1+2i$ lies inside the circle $C: |z+1-i| = 2$. Therefore, $f(z)$ is analytic within C except at this pole.

$$\begin{aligned} \therefore \operatorname{Res} f(-1+2i) &= \lim_{z \rightarrow -1+2i} [(z - (-1+2i)) f(z)] = \lim_{z \rightarrow -1+2i} \frac{(z+1-2i)(z-3)}{z^2+2z+5} \\ &= \lim_{z \rightarrow -1+2i} \frac{z-3}{z+1+2i} = \frac{-4+2i}{4i} = i+1/2. \end{aligned}$$

Hence by residue theorem $\oint_C f(z) dz = 2\pi i \operatorname{Res} f(-1+2i) = 2\pi i(i+1/2) = \pi(i-2)$.

(iii) Here only the pole $z = -1-2i$ lies inside the circle $C: |z+1+i| = 2$. Therefore, $f(z)$ is analytic within C except at this pole.

$$\begin{aligned} \therefore \operatorname{Res} f(-1-2i) &= \lim_{z \rightarrow -1-2i} \frac{(z+1+2i)(z-3)}{z^2+2z+5} \\ &= \lim_{z \rightarrow -1-2i} \frac{z-3}{z+1-2i} = \frac{-4-2i}{-4i} = \frac{1}{2} - i \end{aligned}$$

Hence by residue theorem, $\oint_C f(z) dz = 2\pi i \operatorname{Res} f(-1-2i) = 2\pi i(\frac{1}{2} - i) = \pi(2+i)$.

Example 20.36. Evaluate $\oint_C \frac{e^z}{\cos \pi z} dz$, where C is the unit circle $|z| = 1$.

(Rohtak, 2006)

Solution. $f(z) = e^z/\cos \pi z$ has simple poles at $z = \pm 1/2, \pm 3/2, \pm 5/2, \dots$

Out of these only the poles at $z = 1/2$ and $z = -1/2$ lie inside the given circle $|z| = 1$.

$$\therefore \operatorname{Res} f(1/2) = \lim_{z \rightarrow 1/2} \left[\left(z - \frac{1}{2} \right) f(z) \right] = \lim_{z \rightarrow 1/2} \left\{ \frac{\left(z - \frac{1}{2} \right) e^z}{\cos \pi z} \right\} \quad \left[\frac{0}{0} \text{ form} \right]$$

$$= \lim_{z \rightarrow 1/2} \frac{e^z + \left(z - \frac{1}{2}\right)e^z}{-\pi \sin \pi z} = \frac{e^{1/2}}{-\pi}$$

and $\text{Res } f(-1/2) = \lim_{z \rightarrow -1/2} \left\{ \frac{\left(z + \frac{1}{2}\right)e^z}{\cos \pi z} \right\} \quad \left[\frac{0}{0} \text{ form} \right]$

$$= \lim_{z \rightarrow -1/2} \frac{e^z + \left(z + \frac{1}{2}\right)e^z}{-\pi \sin \pi z} = \frac{e^{-1/2}}{\pi}$$

Hence $\oint_C \frac{e^z}{\cos \pi z} dz = 2\pi i \left(\text{Res } f\left(\frac{1}{2}\right) + \text{Res } f\left(-\frac{1}{2}\right) \right)$

$$= 2\pi i \left(-\frac{e^{1/2}}{\pi} + \frac{e^{-1/2}}{\pi} \right) = -4i \left(\frac{e^{1/2} - e^{-1/2}}{2} \right) = -4i \sinh \frac{1}{2}.$$

Example 20.37. Evaluate $\oint_C \tan z \, dz$ where C is the circle $|z| = 2$.

(V.T.U., 2010 S)

Solution. The poles of $f(z) = \sin z / \cos z$ are given by $\cos z = 0$ i.e. $z = \pm \pi/2, \pm 3\pi/2, \pm 5\pi/2, \dots$
Of these poles, $z = \pi/2$, and $-\pi/2$ only are within the given circle.

$$\therefore \text{Res } f(\pi/2) = \lim_{z \rightarrow \pi/2} \frac{\sin z}{\frac{d}{dz}(\cos z)} = \lim_{z \rightarrow \pi/2} \left(\frac{\sin z}{-\sin z} \right) = -1 \quad [\text{By } \S 20.19 (2)]$$

Similarly $\text{Res } f(-\pi/2) = \lim_{z \rightarrow -\pi/2} \frac{\sin z}{\frac{d}{dz}(\cos z)} = -1.$

Hence by residue theorem,

$$\oint_C f(z) dz = 2\pi i \{ \text{Res } f(\pi/2) + \text{Res } f(-\pi/2) \} = 2\pi i (-1 - 1) = -4\pi i.$$

Example 20.38. Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$, where C is the circle $|z| = 3$.

(V.T.U., 2010 ; Anna, 2003 S ; U.P.T.U., 2002)

Solution. $f(z) = \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)}$

is analytic within the circle $|z| = 3$ excepting the poles $z = 1$ and $z = 2$.

Since $z = 1$ is a pole of order 2.

$$\therefore \text{Res } f(1) = \frac{1}{1!} \left[\frac{d}{dz} \{ (z-1)^2 f(z) \} \right]_{z=1} = \left[\frac{d}{dz} \left(\frac{\sin \pi z^2 + \cos \pi z^2}{(z-2)} \right) \right]_{z=1}$$

$$= \left[\frac{(z-2)(2\pi z \cos \pi z^2 - 2\pi z \sin \pi z^2) - (\sin \pi z^2 + \cos \pi z^2)}{(z-2)^2} \right]_{z=1}$$

$$= (-1)(-2\pi) - (-1) = 2\pi + 1$$

Also $\text{Res } f(2) = \lim_{z \rightarrow 2} [(z-2) f(z)] = \lim_{z \rightarrow 2} \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2} = 1$

Hence by residue theorem,

$$\oint_C f(z) dz = 2\pi i [\text{Res } f(1) + \text{Res } f(2)] = 2\pi i (2\pi + 1 + 1) = 4\pi(\pi + 1)i.$$

PROBLEMS 20.8

- Expand $f(z) = 1/[z^2(z-i)]$ as a Laurent's series about i and hence find the residue thereat. (J.N.T.U., 2003)
- Find the residue of (i) $ze^z/(z-1)^3$ at its pole. (P.T.U., 2009 S)
(ii) $z^2/(z^2+a^2)$ at $z=ai$.
- Determine the poles of the following functions and the residue at each pole :
(i) $\frac{z^2+1}{z^3-2z}$ (ii) $\frac{z^2-2z}{(z+1)^2(z^2+1)}$ (J.N.T.U., 2005) (iii) $\frac{2z+4}{(z+1)(z^2+1)}$ (J.N.T.U., 2006)
- Find the residues of the following functions at each pole.
(i) $(1-e^{2z})/z^4$ (ii) $ze^{iz}/(z^2+1)$ (P.T.U., 2010) (iii) $\cot z$.
- $\oint_C \frac{z^2+4}{(z-2)(z+3)} dz$, where C is (i) $|z+1|=2$ (ii) $|z-2|=2$. (Mumbai, 2006)
- Evaluate the following integrals :
(i) $\oint_C \frac{e^{2z} dz}{(z+2)(z+4)(z+7)}$ for C as circle $|z|=3$. (V.T.U., 2009)
(ii) $\oint_C \frac{4z^2-4z+1}{(z-2)(4+z^2)} dz$, $C: |z|=1$
(iii) $\oint_C \frac{3z^2+z+1}{(z^2-1)(z+3)} dz$, $C: |z|=2$. (U.P.T.U., 2004)
- Evaluate
(i) $\int_C \frac{2z+1}{(2z-1)^2} dz$, where C is $|z|=1$ (ii) $\oint_C \frac{z+4}{z^2+2z+5} dz$, where C is $|z+1-i|=2$
(iii) $\int_C \frac{z^2-2z}{(z+1)^2(z^2+4)} dz$, where C is the circle $|z|=10$. (U.P.T.U., 2009)
- Evaluate :
(i) $\oint_C \frac{z dz}{(z-1)(z-2)^2}$, $C: |z-2|=\frac{1}{2}$. (Madras, 2006)
(ii) $\oint_C \frac{3z^2+2}{(z-1)(z^2+9)} dz$, $C: |z-2|=2$. (Rohtak, 2005)
(iii) $\oint_C \frac{dz}{(z^2+4)^2}$, $C: |z-i|=2$. (Hissar, 2007 ; Anna, 2003 S ; Osmania, 2003)
- Evaluate :
(i) $\oint_C \frac{e^{-z}}{z^2} dz$, $C: |z|=1$. (ii) $\oint_C z^2 e^{1/z}, C: |z|=1$.
(iii) $\oint_C \frac{e^z dz}{z^2+4}$, $C: |z-i|=2$. (V.T.U., 2006) (iv) $\oint_C \frac{e^{2z} dz}{(z+1)^4}$, $C: |z|=2$.
- Evaluate the following integrals : (i) $\oint_C \frac{\sin^6 z}{(z-\pi/6)^3} dz$, $C: |z|=1$
(ii) $\oint_C \frac{z \sec z}{(1-z)^2} dz$, $C: |z|=3$ (iii) $\oint_C \frac{z \cos z}{(z-\pi/2)^3} dz$, $C: |z-1|=1$. (V.T.U., 2007)
- Evaluate $\oint_C \frac{dz}{\sinh 2z}$ where C is the circle $|z|=2$. (Marathwada, 2008)
- Obtain Laurent's expansion for the function $f(z) = 1/z^2 \sinh z$ and evaluate
 $\oint_C \frac{z}{z^2 \sinh z}$, where C is the circle $|z-1|=2$. (J.N.T.U., 2005)

20.20 EVALUATION OF REAL DEFINITE INTEGRALS

Many important definite integrals can be evaluated by applying the Residue theorem to properly chosen integrals. The contours chosen will consist of straight lines and circular arcs.

(a) **Integration around the unit circle.** An integral of the type $\int_0^{2\pi} f(\sin \theta, \cos \theta) d\theta$, where the integrand is a rational function of $\sin \theta$ and $\cos \theta$ can be evaluated by writing $e^{i\theta} = z$.

Since $\sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right)$ and $\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$, then integral takes the form $\int_C f(z) dz$, where $f(z)$ is a rational function of z and C is a unit circle $|z| = 1$.

Hence the integral is equal to $2\pi i$ times the sum of the residues at those poles of $f(z)$ which are within C .

Example 20.39. Show that

$$\int_0^{2\pi} \frac{\cos 2\theta d\theta}{1 - 2a \cos \theta + a^2} = \frac{2\pi a^2}{1 - a^2}, \quad (a^2 < 1). \quad (\text{Bhopal, 2009 ; Rohtak, 2003})$$

Solution. Putting $z = e^{i\theta}$, $d\theta = dz/iz$, $\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$ and $\cos 2\theta = \frac{1}{2} (e^{2i\theta} + e^{-2i\theta}) = \frac{1}{2} (z^2 + 1/z^2)$

$\therefore$ the given integral

$$\begin{aligned} I &= \int_C \frac{\frac{1}{2}(z^2 + 1/z^2)}{1 - a(z + 1/z) + a^2} \cdot \frac{dz}{iz} = \frac{1}{2i} \int_C \frac{(z^4 + 1) dz}{z^2(z - az^2 - a + a^2z)} \\ &= \frac{1}{2i} \int_C \frac{(z^4 + 1) dz}{z^2(z - a)(1 - az)} = \int_C f(z) dz \quad \text{where } C \text{ is the unit circle } |z| = 1. \end{aligned}$$

Now $f(z)$ has simple poles at $z = a$, $1/a$ and the second order pole at $z = 0$, of which the poles at $z = 0$ and $z = a$ lie within the unit circle.

$$\therefore \text{Res } f(a) = \lim_{z \rightarrow a} [(z - a) f(z)] = \frac{1}{2i} \lim_{z \rightarrow a} \left[\frac{z^4 + 1}{z^2(1 - az)} \right] = \frac{a^4 + 1}{2ia^2(1 - a^2)}$$

$$\begin{aligned} \text{and } \text{Res } f(0) &= \lim_{z \rightarrow 0} \frac{d}{dz} [z^2 f(z)] = \frac{1}{2i} \lim_{z \rightarrow 0} \frac{d}{dz} \left[\frac{z^4 + 1}{z - az^2 - a + a^2z} \right] \\ &= \frac{1}{2i} \lim_{z \rightarrow 0} \frac{(z - az^2 - a + a^2z)(4z^3) - (z^4 + 1)(1 - 2az + a^2)}{(z - az^2 - a + a^2z)^2} = -\frac{1 + a^2}{2ia^2} \end{aligned}$$

$$\text{Hence } I = 2\pi i [\text{Res } f(a) + \text{Res } f(0)] = 2\pi i \left[\frac{a^4 + 1}{2ia^2(1 - a^2)} - \frac{1 + a^2}{2ia^2} \right] = \frac{2\pi a^2}{1 - a^2}.$$

Example 20.40. By integrating around a unit circle, evaluate $\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta$.

(S.V.T.U., 2009 ; U.P.T.U., 2009 ; Madras, 2003)

Solution. Putting $z = e^{i\theta}$, $d\theta = dz/iz$, $\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$

$$\text{and } \cos 3\theta = \frac{1}{2} (e^{3i\theta} + e^{-3i\theta}) = \frac{1}{2} (z^3 + 1/z^3).$$

$$\begin{aligned} \therefore \text{ the given integral } I &= \int_C \frac{\frac{1}{2}(z^3 + 1/z^3)}{5 - 2(z + 1/z)} \cdot \frac{dz}{iz} \\ &= -\frac{1}{2i} \int_C \frac{z^6 + 1}{z^3(2z^2 - 5z + 2)} dz = -\frac{1}{2i} \int_C \frac{(z^6 + 1) dz}{z^3(2z - 1)(z - 2)} \end{aligned}$$

$$= -\frac{1}{2i} \int_C f(z) dz, \quad \text{where } C \text{ is the unit circle } |z| = 1.$$

Now $f(z)$ has a pole of order 3 at $z = 0$ and simple poles at $z = \frac{1}{2}$ and $z = 2$. Of these only $z = 0$ and $z = 1/2$ lie within the unit circle.

$$\begin{aligned} \therefore \text{Res } f(1/2) &= \lim_{z \rightarrow 1/2} \frac{(z - 1/2)(z^6 + 1)}{(2z - 1)(z - 2)} = \lim_{z \rightarrow 1/2} \left\{ \frac{z^6 + 1}{2z^3(z - 2)} \right\} = -\frac{65}{24} \\ \text{Res } f(0) &= \frac{1}{(n-1)!} \left\{ \frac{d^{n-1}}{dz^{n-1}} [(z-0)^n f(z)] \right\}_{z=0} \quad \text{where } n = 3 \\ &= \frac{1}{2} \left\{ \frac{d^2}{dz^2} \left(\frac{z^6 + 1}{2z^2 - 5z + 2} \right) \right\}_{z=0} = \frac{d}{dz} \left[\frac{(2z^2 - 5z + 2)6z^5 - (z^6 + 1)(4z - 5)}{2(2z^2 - 5z + 2)^2} \right]_{z=0} \\ &= \left\{ \frac{d}{dz} \left[\frac{8z^7 - 25z^6 + 12z^5 - 4z + 5}{2(2z^2 - 5z + 2)^2} \right] \right\}_{z=0} \\ &= \left[\frac{(2z^2 - 5z + 2)^2 (56z^6 - 150z^5 + 60z^4 - 4) - (8z^7 - 25z^6 + 12z^5 - 4z + 5) 2(2z^2 - 5z + 2)(4z - 5)}{2(2z^2 - 5z + 2)^4} \right]_{z=0} \\ &= \frac{4(-4) - 5(-20)}{2 \times 16} = \frac{84}{32} = \frac{21}{8} \end{aligned}$$

$$\text{Hence } I = \frac{-1}{2i} \{2\pi i [\text{Res } f(1/2) + \text{Res } f(0)]\} = -\pi \left[-\frac{65}{24} + \frac{21}{8} \right] = -\pi \left(-\frac{1}{12} \right) = \frac{\pi}{12}.$$

(b) **Integration around a small semi-circle.** To evaluate $\int_{-\infty}^{\infty} f(x) dx$, we consider $\int_C f(z) dz$, where C is the contour consisting of the semi-circle $C_R: |z| = R$, together with the diameter that closes it.

Supposing that $f(z)$ has no singular point on the real axis, we have, by the Residue theorem,

$$\int_{C_R} f(z) dz + \int_{-R}^R f(x) dx = 2\pi i \sum \text{Res } f(a).$$

Finally making R tend to ∞ , we find the value of $\int_{-\infty}^{\infty} f(x) dx$, provided $\int_{C_R} f(z) dz \rightarrow 0$ as $R \rightarrow \infty$.

Example 20.41. Evaluate $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2 + 1)(x^2 + 4)}$.

(U.P.T.U., 2008)

Solution. Consider $\int_C \frac{z^2 dz}{(z^2 + 1)(z^2 + 4)} = \int_C f(z) dz$

where C is the contour consisting of the semi-circle C_R of radius R together with the part of the real axis from $-R$ to R as shown in Fig. 20.23.

The integrand has simple poles at $z = \pm i, z = \pm 2i$ of which $z = i, 2i$ only lie inside C .

$\therefore$ by the Residue theorem,

$$\begin{aligned} \int_C f(z) dz &= 2\pi i [\text{Res } f(i) + \text{Res } f(2i)] \\ &= 2\pi i \left[\lim_{z \rightarrow i} (z - i) f(z) + \lim_{z \rightarrow 2i} (z - 2i) f(z) \right] \\ &= 2\pi i \left[\frac{i^2}{2i(i^2 + 4)} + \frac{4i^2}{(4i^2 + 1)(4i)} \right] = 2\pi i \left(\frac{i}{6} - \frac{i}{3} \right) = \frac{\pi}{3} \end{aligned} \quad \dots(ii)$$

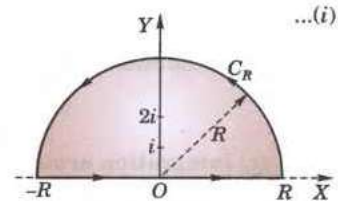


Fig. 20.23

Also
$$\int_C f(z) dz = \int_{-R}^R f(x) dx + \int_{C_R} f(z) dz \quad \dots(iii)$$

Now let $R \rightarrow \infty$, so as to show that the second integral in (iii) vanishes. For any point on C_R as $|z| \rightarrow \infty$

$$f(z) = \frac{1}{z^2} \cdot \frac{1}{(1+z^{-2})(1+4z^{-2})}$$

decreases as $1/z^2$ and tends to zero whereas the length of C_R increases with z .

Consequently,
$$\lim_{|z| \rightarrow \infty} \int_{C_R} f(z) dz = 0.$$

Hence from (i), (ii) and (iii), we get
$$\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+1)(x^2+4)} = \frac{\pi}{3}.$$

Example 20.42. Evaluate $\int_0^{\infty} \frac{\cos ax}{x^2+1} dx$.

(U.P.T.U., 2006; Delhi, 2002)

Solution. Consider
$$\int_C \frac{e^{iaz}}{z^2+1} dz = \int_C f(z) dz$$

where C is the contour consisting of the semi-circle C_R of radius R together with the part of the real axis from $-R$ to R as shown in Fig. 20.23.

The integrand has simple poles at $z = i$ and $z = -i$, of which $z = i$ only lies inside C .

$\therefore$ by Residue theorem,
$$\begin{aligned} \int_C f(z) dz &= 2\pi i \operatorname{Res} f(i) = 2\pi i \lim_{z \rightarrow i} [(z-i) f(z)] \\ &= 2\pi i \lim_{z \rightarrow i} \frac{(z-i) e^{iaz}}{z^2+1} = 2\pi i \lim_{z \rightarrow i} \frac{e^{iaz}}{z+i} = 2\pi i \frac{e^{-a}}{2i} = \pi e^{-a} \end{aligned} \quad \dots(i)$$

Also
$$\int_C f(z) dz = \int_{-R}^R f(x) dx + \int_{C_R} f(z) dz \quad \dots(ii)$$

Now $|z| = R$ on C_R and $|z^2+1| \geq R^2-1$.

Also $|e^{iaz}| = |e^{ia(x+iy)}| = |e^{iax} \cdot e^{-ay}| = e^{-ay} < 1 \quad [\because y > 0]$

$\therefore \left| \frac{e^{iaz}}{z^2+1} \right| = |e^{iaz}| \cdot \frac{1}{|z^2+1|} < 1 \cdot \frac{1}{R^2-1}$

Thus
$$\int_{C_R} f(z) dz = \left| \int_{C_R} \frac{e^{iaz}}{z^2+1} dz \right| < \int_{C_R} \frac{1}{R^2-1} |dz| < \frac{\pi R}{R^2-1} \text{ which } \rightarrow 0 \text{ as } R \rightarrow \infty. \quad \dots(iii)$$

Hence from (i), (ii) and (iii), we get

$$\pi e^{-a} = \int_{-\infty}^{\infty} f(x) dx + 0 \quad \text{or} \quad \int_{-\infty}^{\infty} \frac{e^{iax}}{x^2+1} dx = \pi e^{-a}$$

Equating real parts from both sides, we obtain

$$\int_{-\infty}^{\infty} \frac{\cos ax}{x^2+1} dx = \pi e^{-a}$$

Since $\cos ax/(x^2+1)$ is an even function of x , we have

$$2 \int_0^{\infty} \frac{\cos ax}{x^2+1} dx = \pi e^{-a} \quad \text{or} \quad \int_0^{\infty} \frac{\cos ax}{x^2+1} dx = \frac{\pi}{2} e^{-a}.$$

(c) Integration around rectangular contours

Example 20.43. Evaluate $\int_{-\infty}^{\infty} \frac{e^{ax}}{e^x+1} dx$.

Solution. Consider
$$\int_C \frac{e^{az}}{e^z+1} dz = \int_C f(z) dz$$
 where C is the rectangle $ABCD$ with vertices at $(R, 0)$, $(R, 2\pi)$, $(-R, 2\pi)$ and $(-R, 0)$, R being positive (Fig. 20.24).

$f(z)$ has finite poles given by

$$e^z = -1 = e^{(2n+1)\pi i}$$

or

$$z = (2n+1)\pi i, \text{ where } n = 0, \pm 1, \pm 2, \pm 3, \dots$$

The only pole inside the rectangle is $z = \pi i$.

$\therefore$ by Residue theorem,

$$\begin{aligned} \int_C f(z) dz &= 2\pi i \operatorname{Res} f(\pi i) \\ &= 2\pi i \left[e^{az} \frac{d}{dz} (e^z + 1) \right]_{z=\pi i} \\ &= 2\pi i e^{a\pi i} / e^{\pi i} = -2\pi i e^{a\pi i} \quad [\because e^{\pi i} = -1] \end{aligned} \quad \dots(i)$$

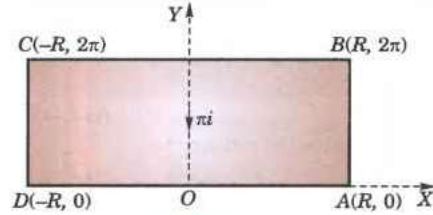


Fig. 20.24

$$\begin{aligned} \text{Also } \int_C f(z) dz &= \int_{AB} f(z) dz + \int_{BC} f(z) dz + \int_{CD} f(z) dz + \int_{DA} f(z) dz \\ &= \int_0^{2\pi} f(R+iy) idy + \int_R^{-R} f(x+2\pi i) dx + \int_{2\pi}^0 f(-R+iy) idy + \int_{-R}^R f(x) dx \\ &[\because z = R+iy \text{ along } AB, z = x+2\pi i \text{ along } BC, z = -R+iy \text{ along } CD \text{ and } z = x \text{ along } DA.] \end{aligned}$$

$$\text{or } \int_C f(z) dz = i \int_0^{2\pi} \frac{e^{a(R+iy)}}{e^{R+iy} + 1} dy - \int_{-R}^R \frac{e^{a(x+2\pi i)}}{e^{x+2\pi i} + 1} dx - i \int_0^{2\pi} \frac{e^{a(-R+iy)}}{e^{-R+iy} + 1} dy + \int_{-R}^R \frac{e^{ax}}{e^x + 1} dx \quad \dots(ii)$$

Now for any two complex numbers z_1, z_2

$$|z_1| \geq |z_2|, \text{ we have } |z_1 + z_2| \geq |z_1| - |z_2|$$

so that $|e^{R+iy} + 1| \geq e^R - 1$. Also $|e^{a(R+iy)}| = e^{aR}$

$\therefore$ for the integrand of first integral in (ii), we have

$$\left| \frac{e^{a(R+iy)}}{e^{R+iy} + 1} \right| \leq \frac{e^{aR}}{e^R - 1} \text{ which } \rightarrow 0 \text{ as } R \rightarrow \infty. \quad [\because a > 1]$$

Similarly, for the integrand of the third integral in (ii), we get

$$\left| \frac{e^{a(-R+iy)}}{e^{-R+iy} + 1} \right| \leq \frac{e^{-aR}}{1 - e^{-R}} \text{ which also } \rightarrow 0 \text{ as } R \rightarrow \infty. \quad [\because a < 0]$$

Hence as $R \rightarrow \infty$, since the first and third integrals in (ii) approach zero, we get

$$\int_C f(z) dz = -e^{2a\pi i} \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx + \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx = (1 - e^{2a\pi i}) \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx \quad \dots(iii)$$

Thus from (i) and (iii), we obtain $(1 - e^{2a\pi i}) \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx = -2\pi i e^{a\pi i}$

$\therefore$ equating real parts, we get $\int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx = \frac{\pi}{\sin a\pi}$.

Example 20.44. Show that $\int_0^{\infty} e^{-x^2} \cos 2mx \, dx = \frac{1}{2} \sqrt{\pi} e^{-m^2}$.

Solution. Integrate $f(z) = e^{-z^2}$ along the rectangle ABCDA having vertices $A(-l)$, $B(l)$, $C(l+im)$, $D(-l+im)$ (Fig. 20.25). $f(z)$ has no poles inside this contour. As such

$$\int_{AB} f(z) dz + \int_{BC} f(z) dz + \int_{CD} f(z) dz + \int_{DA} f(z) dz = 0 \quad \dots(i)$$

On $AB : z = x$, on $BC : z = l + iy$, on $CD : z = x + im$ and on $DA : z = -l + iy$.

Therefore, (i) becomes

$$\int_{-l}^l e^{-x^2} dx + \int_0^m e^{-(l+iy)^2} idy + \int_l^{-l} e^{-(x+im)^2} dx + \int_m^0 e^{-(-l+iy)^2} dy = 0$$

or

$$\begin{aligned} \int_{-l}^l e^{-x^2} dx - \int_{-l}^l e^{-x^2 - 2imx + m^2} dx + \int_0^m e^{-l^2 - 2ily + y^2} \cdot idy \\ - \int_0^m e^{-l^2 + 2ily + y^2} \cdot idy = 0 \end{aligned} \quad \dots(ii)$$

Now let $l \rightarrow \infty$. Then the last two integrals

$$= ie^{-l^2} \int_0^m e^{y^2} (e^{-2ily} - e^{2ily}) dy = 2e^{-l^2} \int_0^m e^{y^2} \sin 2ly dy \rightarrow 0$$

$\therefore$ As $l \rightarrow \infty$, $e^{-l^2} \rightarrow 0$ and $\sin 2ly$ is finite

Hence (ii) reduces to

$$\int_{-\infty}^{\infty} e^{-x^2} dx - e^{m^2} \int_{-\infty}^{\infty} e^{-x^2} (\cos 2mx - i \sin 2mx) dx = 0$$

Equating real parts, we get

$$e^{m^2} \int_{-\infty}^{\infty} e^{-x^2} \cos 2mx dx = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

[See p. 289]

or

$$\int_0^{\infty} e^{-x^2} \cos 2mx dx = \frac{1}{2} \sqrt{\pi} e^{-m^2}$$

(d) **Indenting the contours having poles on the real axis.** So far we have considered such cases in which there is no pole on the real axis. When the integrand has a simple pole on the real axis, we delete it from the region by indenting the contour (i.e., by drawing a small semi-circle having the pole for the centre). The method will be clear from the following example.

Example 20.45. Evaluate $\int_0^{\infty} \frac{\sin mx}{x} dx$, when $m > 0$.

(U.P.T.U., 2007)

Solution. Consider the integral $\int_C \frac{e^{miz}}{z} dz = \int_C f(z) dz$ where C consists of

- (i) the real axis from r to R ,
- (ii) the upper half of the circle $C_R: |z| = R$,
- (iii) the real axis $-R$ to $-r$,
- (iv) the upper half of the circle $C_r: |z| = r$ (Fig. 20.26).

Since $f(z)$ has no singularity inside C (its only singular point being a simple pole at $z = 0$ which has been deleted by drawing C_r), we have by Cauchy's theorem :

$$\int_r^R f(x) dx + \int_{C_R} f(z) dz + \int_{-R}^{-r} f(x) dx + \int_{C_r} f(z) dz = 0 \quad \dots(i)$$

$$\begin{aligned} \text{Now} \quad \int_{C_R} f(z) dz &= \int_0^{\pi} \frac{e^{imR(\cos \theta + i \sin \theta)}}{Re^{i\theta}} \cdot Rie^{i\theta} d\theta \\ &= i \int_0^{\pi} e^{imR(\cos \theta + i \sin \theta)} d\theta \end{aligned}$$

$$\text{Since} \quad |e^{imR(\cos \theta + i \sin \theta)}| = |e^{-mR \sin \theta + imR \cos \theta}| = e^{-mR \sin \theta}$$

$$\begin{aligned} \therefore \quad \left| \int_{C_R} f(z) dz \right| &\leq \int_0^{\pi} e^{-mR \sin \theta} d\theta = 2 \int_0^{\pi/2} e^{-mR \sin \theta} d\theta \\ &= 2 \int_0^{\pi/2} e^{-2mR\theta/\pi} d\theta \end{aligned}$$

$[\because \text{ for } 0 \leq \theta \leq \pi/2, \sin \theta/\theta \geq 2/\pi]$

$$= \frac{\pi}{mR} (1 - e^{-mR}) \text{ which } \rightarrow 0 \text{ as } R \rightarrow \infty,$$

$$\text{Also} \quad \int_{C_r} f(z) dz = i \int_{\pi}^0 e^{imr(\cos \theta + i \sin \theta)} d\theta \rightarrow i \int_{\pi}^0 d\theta \text{ i.e., } -i\pi \text{ as } r \rightarrow 0.$$

$$\text{Hence as } r \rightarrow 0 \text{ and } R \rightarrow \infty, \text{ we get from (i) } \int_0^{\infty} f(x) dx + 0 + \int_{-\infty}^0 f(x) dx - i\pi = 0$$

$$\text{or} \quad \int_{-\infty}^{\infty} f(x) dx = i\pi \text{ i.e., } \int_{-\infty}^{\infty} \frac{e^{imx}}{x} dx = i\pi \quad \dots(ii)$$

Equating imaginary parts from both sides,

$$\int_{-\infty}^{\infty} \frac{\sin mx}{x} dx = \pi. \text{ Hence } \int_0^{\infty} \frac{\sin mx}{x} dx = \frac{\pi}{2}.$$

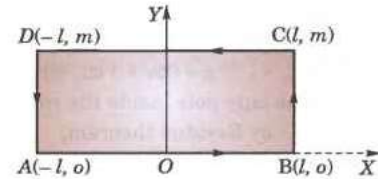


Fig. 20.25

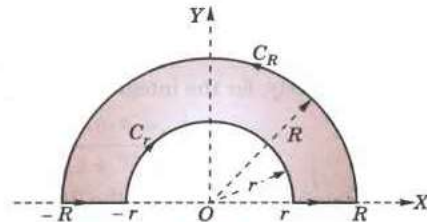


Fig. 20.26

Obs. Equating real parts from both sides of (ii), we get

$$\int_{-\infty}^{\infty} \frac{\cos mx}{x} dx = 0.$$

Example 20.46. Show that $\int_0^{\infty} \frac{x^{p-1}}{1+x} dx = \frac{\pi}{\sin p\pi}$, $0 < p < 1$.

Solution. Integrate $f(z) = \frac{z^{p-1}}{1+z}$ along the contour consisting of the circles α and γ of radii a and R and the lines AB and FG along x -axis (Fig. 20.27). There is a simple pole at $z = -1$ which is within the contour.

$$\therefore \text{Res } f(-1) = \lim_{z \rightarrow -1} (1+z) \cdot \frac{z^{p-1}}{1+z} = \lim_{z \rightarrow -1} z^{p-1} = (-1)^{p-1} = e^{i\pi(p-1)}$$

$$\text{Thus } \int_{AB} f(z) dz + \int_{\gamma} f(z) dz + \int_{FG} f(z) dz + \int_{\alpha} f(z) dz = 2\pi i e^{i\pi(p-1)} \quad \dots(i)$$

On $AB : z = x$ and on $FG : z = xe^{2\pi i}$

$$\begin{aligned} \therefore \int_{AB} f(z) dz + \int_{FG} f(z) dz &= \int_a^R \frac{x^{p-1}}{1+x} dx + \int_R^a \frac{(xe^{2\pi i})^{p-1}}{1+xe^{2\pi i}} dx e^{2\pi i} \\ &= \int_a^R \frac{x^{p-1}}{1+x} [1 - e^{2\pi i(p-1)}] dx \end{aligned}$$

On the circle $\gamma : z = Re^{i\theta}$. So

$$\int_{\gamma} f(z) dz = \int_0^{2\pi} \frac{(Re^{i\theta})^{p-1}}{1+Re^{i\theta}} Re^{i\theta} i d\theta$$

For large R , the integrand is of the order $\frac{R^{p-1}}{1+R}$ i.e.

R^{p-1} which tends to zero as $R \rightarrow \infty$.

($\because p < 1$)

Hence $\int_{\gamma} f(z) dz \rightarrow 0$ as $R \rightarrow \infty$

On the circle $\alpha : z = ae^{i\theta}$. So

$$\int_{\alpha} f(z) dz = \int_{2\pi}^0 \frac{(ae^{i\theta})^{p-1}}{1+ae^{i\theta}} ae^{i\theta} i d\theta$$

For small a , the integrand is of the order a^p which tends to zero as $a \rightarrow 0$. ($\because p > 0$)

Thus on taking limits as $a \rightarrow 0$ and $R \rightarrow \infty$, (i) gives

$$\int_0^{\infty} \frac{x^{p-1}}{1+x} (1 - e^{2\pi i(p-1)}) dx = 2\pi i e^{i\pi(p-1)}$$

or

$$\int_0^{\infty} \frac{x^{p-1}}{1+x} dx = \frac{2\pi i e^{i\pi(p-1)}}{1 - e^{2\pi i(p-1)}} = \frac{2\pi i e^{i\pi p} (-1)}{1 - e^{2i\pi p} (1)} = \frac{2i \cdot \pi}{e^{i\pi p} - e^{-i\pi p}} = \frac{\pi}{\sin p\pi}.$$

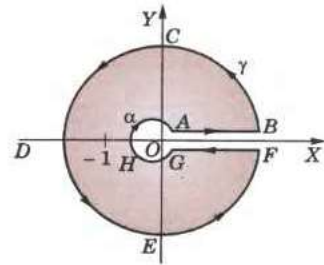


Fig. 20.27

Example 20.47. Prove that $\int_0^{\infty} \sin x^2 dx = \int_0^{\infty} \cos x^2 dx = \frac{1}{2} \sqrt{\left(\frac{\pi}{2}\right)}$.

(Osmania, 2003)

Solution. Consider $\int_C e^{-z^2} dz$ where C consists of the real axis from O to A , part of circle AB of radius R and the line $\theta = \frac{\pi}{4}$. (Fig. 20.28).

e^{-z^2} has no singularity within C .

$$\therefore \int_{OA} e^{-z^2} dz + \int_{AB} e^{-z^2} dz + \int_{BO} e^{-z^2} dz = 0$$

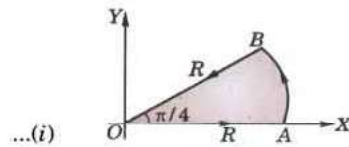


Fig. 20.28

On $OA : z = x$, $\therefore \int_{OA} e^{-z^2} dz = \int_0^R e^{-x^2} dx \rightarrow \sqrt{\pi}/2$ as $R \rightarrow \infty$ [See p. 289]

On $AB : z = Re^{i\theta}$,

$$\therefore \int_{AB} e^{-z^2} dz = \int_0^{\pi/4} e^{-R^2(\cos 2\theta + i \sin 2\theta)} \cdot Re^{i\theta} \cdot i d\theta \rightarrow 0 \quad \text{as } R \rightarrow \infty.$$

[$\because$ integrand $\rightarrow 0$ as $R \rightarrow \infty$]

On $BO : z = re^{i\pi/4}$ and $z^2 = r^2 e^{i\pi/2} = ir^2$

$$\begin{aligned} \therefore \int_{BO} e^{-z^2} dz &= \int_R^0 e^{-ir^2} \cdot e^{i\pi/4} dr = - \int_0^R e^{-ix^2} \frac{1+i}{\sqrt{2}} dx \\ &\rightarrow - \int_0^\infty (\cos x^2 - i \sin x^2) \frac{1+i}{\sqrt{2}} dx \quad \text{when } R \rightarrow \infty \end{aligned}$$

Substituting these in (i), we get

$$\frac{1}{2}\sqrt{\pi} + 0 - \int_0^\infty (\cos x^2 - i \sin x^2) \left(\frac{1+i}{\sqrt{2}} \right) dx = 0$$

Equating real and imaginary parts, we obtain

$$\int_0^\infty (\cos x^2 + \sin x^2) dx = \frac{1}{2}\sqrt{(2\pi)} \quad \text{and} \quad \int_0^\infty (\cos x^2 - \sin x^2) dx = 0$$

Hence $\int_0^\infty \sin x^2 dx = \int_0^\infty \cos x^2 dx = \frac{1}{2}\sqrt{\left(\frac{\pi}{2}\right)}.$

PROBLEMS 20.9

Apply the calculus of residues, to prove that

- $\int_0^{2\pi} \frac{d\theta}{1 - 2p \sin \theta + p^2} = \frac{2\pi}{1 - p^2} \quad (0 < p < 1).$ (Hissar, 2007; Mumbai, 2006; Kerala, 2005)
- $\int_0^{2\pi} \frac{d\theta}{1 - 2r \cos \theta + r^2} = \frac{\pi}{1 - r^2}.$ (J.N.T.U., 2006; Madras, 2006; Anna, 2003)
- $\int_0^{2\pi} \frac{d\theta}{a + \cos \theta} = \frac{2\pi}{\sqrt{a^2 - 1}} \quad (a > 1).$ (P.T.U., 2010)
- $\int_0^{2\pi} \frac{\cos 2\theta}{5 + 4 \cos \theta} d\theta = \frac{\pi}{6}.$ (U.P.T.U., 2010)
- $\int_0^{2\pi} \frac{\sin^2 \theta d\theta}{a + b \cos \theta} = \frac{2\pi}{b^2} [a - \sqrt{a^2 - b^2}], \quad (0 < b < a).$ (J.N.T.U., 2003)
- $\int_0^{2\pi} \frac{a d\theta}{a^2 + \sin^2 \theta} = \frac{2\pi}{\sqrt{1 + a^2}}, \quad (a > 0).$ (S.V.T.U., 2009)
- $\int_0^{2\pi} \frac{d\theta}{(5 - 3 \cos \theta)^2} = \frac{5\pi}{32}.$
- $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2 + a^2)(x^2 + b^2)} = \frac{\pi}{a + b} \quad (a, b > 0).$ (P.T.U., 2007; Mumbai, 2006; Anna, 2003)
- $\int_{-\infty}^{\infty} \frac{x^2 - x + 2}{x^4 + 10x^2 + 9} dx = \frac{5\pi}{12}.$ (A.M.I.E.T.E., 2003; Delhi, 2002)
- $\int_{-\infty}^{\infty} \frac{dx}{x^4 + 1} = \frac{\pi}{\sqrt{2}}.$ (J.N.T.U., 2006)
- $\int_{-\infty}^{\infty} \frac{dx}{(1 + x^2)^2} = \frac{\pi}{4}.$ (Madras, 2006; Kerala, 2005)
- $\int_0^{\infty} \frac{dx}{x^6 + 1} = \frac{\pi}{3}.$ (Kerala, 2005)
- $\int_0^{\infty} \frac{x^2}{x^6 + 1} dx = \frac{\pi}{6}.$ (Rohtak, 2006)
- $\int_{-\infty}^{\infty} \frac{\cos mx}{e^x + e^{-x}} dx = \frac{\pi}{2} \operatorname{sech} \frac{m\pi}{2}.$
- $\int_0^{\infty} \frac{1 - \cos x}{x^2} dx = \pi/2.$ (P.T.U., 2005)
- $\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}.$ (Kerala, 2005)
- $\int_{-\infty}^{\infty} \frac{\sin x}{x^2 + 4x + 5} dx = \frac{-\pi \sin 2}{e}.$
- $\int_{-\infty}^{\infty} \frac{\cos x dx}{(x^2 + a^2)(x^2 + b^2)} = \frac{\pi}{a^2 - b^2} \left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a} \right) \quad (a > b > 0).$
- $\int_0^{\infty} \frac{x \sin x}{x^2 + a^2} dx = \frac{\pi}{2} e^{-a}.$

20.12. OBJECTIVE TYPE OF QUESTIONS

PROBLEMS 20.10

Select the correct answer or fill up the blanks in each of the following questions :

- The only function that is analytic from the following is
 (i) $f(z) = \sin z$ (ii) $f(z) = \bar{z}$ (iii) $f(z) = \operatorname{Im}(z)$ (iv) $R(iz)$.
- If $f(z) = u(x, y) + iv(x, y)$ is analytic, then $f'(z) =$
 (i) $\frac{\partial u}{\partial x} - i \frac{\partial v}{\partial x}$ (ii) $\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$ (iii) $\frac{\partial v}{\partial y} - i \frac{\partial v}{\partial x}$.
- If $2x - x^2 + ay^2$ is to be harmonic, then a should be
 (a) 1 (b) 2 (c) 3 (d) 0.
- The analytic function which maps the angular region $0 \leq \theta \leq \pi/4$ onto the upper half plane is
 (i) z^2 (ii) $4z$ (iii) z^4 (iv) 2θ .
- An angular domain in the complex plane is defined by $0 < \arg(z) < \pi/4$. The mapping which maps this region onto the left half plane is
 (i) $w = z^4$ (ii) $w = iz^4$ (iii) $w = -z^4$ (iv) $w = -iz^4$.
- The mapping $w = z^2 - 2z - 3$ is
 (i) conformal within $|z| = 1$ (ii) not conformal at $z = 1$
 (iii) not conformal at $z = -1$ and $z = 3$ (iv) conformal everywhere.
- If $z = re^{i\theta}$, then the image of $\theta = \text{constant}$ under the mapping $w(z) = Re^{i\phi} = iz^3$ is
 (i) $\phi = 3\theta$ (ii) $\phi = 3\theta + \pi/2$ (iii) $\phi = 3\theta - \pi/2$ (iv) $\phi = \theta^3$.
- The fixed points of the mapping $w = (5z + 4)/(z + 5)$ are
 (i) 2, 2 (ii) 2, -2 (iii) -2, -2 (iv) -4/5, 5.
- The value of $\int_C (4x^3 dx + 3y^2 z^2 dy + 2y^3 z dz)$ where C is any path joining $A(-1, 1, 0)$ to $B(1, 2, 1)$ is
 (i) 0 (ii) 1 (iii) 8 (iv) -8.
- The value of $\int_C \frac{3z^2 + 7z + 1}{z + 1} dz$ where C is $|z| = 1/2$ is
 (i) $2\pi i$ (ii) 0 (iii) πi (iv) $\pi i/2$.
- The value of $\int_C \frac{3z + 4}{z(2z + 1)} dz$ where C is the circle $|z| = 1$ is
 (i) $2\pi i$ (ii) $3\pi i$ (iii) 4 (iv) -4.
- The residue of a function can be found if the pole is an isolated singularity :
 (i) True (ii) False (iii) Partially false (iv) none of these.
- The value of $\int_C \frac{z dz}{\sin z}$ where $C : |z| = 4$ is
 (i) $2\pi i$ (ii) 0 (iii) $-2\pi i$ (iv) $4\pi i$.
- The value of $\int_C \tanh z dz$, where $C : |z| = 3$, is
 (i) 0 (ii) πi (iii) $2\pi i$ (iv) $4\pi i$.
- The harmonic conjugate of the function $u(x, y) = 2x(1 - y)$ is (U.P.T.U., 2009)
- Harmonic conjugate of $x^3 - 3xy^2$ is
- The curves $u(x, y) = c$ and $v(x, y) = c'$ are orthogonal if
- The value of $\int_0^{1+i} z^2 dz$ along the line $x = y$ is
- Residue of $\frac{\cos z}{z}$ at $z = 0$ is
- The critical point of the transformation $w^2 = (z - a)(z - b)$ is
- Image of $|z + 1| = 1$ under the mapping $w = 1/z$ is
- The poles of $f(z) = (z^3 - 1)/(z^3 + 1)$ are $z =$
- $w = \log z$ is analytic everywhere except at $z =$
- If $f(z) = -\frac{1}{z-1} - 2[1 + (z-1) + (z-1)^2 + \dots]$, then the residue of $f(z)$ at $z = 1$ is
- If $|z| < 1$ then Taylor's series expansion of $\log(1 + z)$ about $z = 0$ is

26. The value of $\int_C \frac{4z^2 + z + 5}{z - 4} dz$ where C is $9x^2 + 4y^2 = 36$, is
27. The value of $\int z^4 e^{1/z} dz$, where C is $|z| = 1$, is
 (i) πi (ii) $\pi i/12$ (iii) $\pi i/60$ (iv) $-\pi i/60$.
28. If $f(z)$ has a pole of order three at $z = a$ $\text{Res}[f(a)] = \dots$
29. The value of $\int_C \frac{e^z dz}{(z-3)^2}$, C being $|z| = 2$, is
30. The CR equations for $f(z) = u(x, y) + iv(x, y)$ to be analytic are
31. If $f(z)$ is analytic in a simply connected domain D and C is any simple closed path then $\int_C f(z) dz = \dots$
32. The harmonic conjugate of $e^x \cos y$ is
33. The value of $\oint_C \cos z dz$ where C is the circle $|z| = 1$, is
34. The singularity of $f(z) = z/(z-2)^3$ is 35. The function $f(z) = \bar{z}$ is analytic at
36. C-R equations for a function to be analytic, in polar form, are
37. If C is the circle $|z - a| = r$, $\int_C (z - a)^n dz$, any integer $n \neq -1$ is
38. A simply connected region is that 39. A holomorphic function is that
40. The poles of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ are at $z = \dots$
41. The cross-ratio of four points z_1, z_2, z_3, z_4 is
42. The value of $\int_C |z| dz$, where C is the contour represented by the straight line from $z = -i$ to $z = i$, is
43. Taylor's series expansion of $\left(\frac{1}{z-2} - \frac{1}{z-1}\right)$ in the region $|z| < 1$, is
44. The invariant points of the transformation $w = (1+z)/(1-z)$ are $z = \dots$
45. The residue at $z = 0$ of $\frac{1+e^z}{z \cos z + \sin z}$ is 46. The transformation $w = Cz$ consists of
47. The residue of $f(z)$ at a pole is 48. The value of $\int_C \frac{1}{z-1} dz$, C being $|z| = 2$, is
49. If C is $|z| = 1/2$, $\int_C \frac{z^2 - z + 1}{z-1} dz = \dots$ 50. Singular points of $\frac{\cos \pi z}{(z-1)(z-2)}$ are
51. Taylor series expansion of $\frac{1}{z-2}$ in $|z| < 1$ is 52. $\text{Lt}_{z \rightarrow \infty} \frac{iz^3 + iz - 1}{(2z+3)(z-1)^2} = \dots$ (P.T.U., 2007)
53. The poles of $\frac{(z-1)^2}{z(z-2)^2}$ are at $z = \dots$ 54. Cauchy's integral theorem states that
55. The critical points of the transformation $w = z + 1/z$ are
56. $\int_C \frac{dz}{2z-3}$, where $|z| = 1$, is 57. The zeroes and singularities of $\frac{z^2+1}{1-z^3}$ are
58. Residue of $\tan z$ at $z = \pi/2$ is 59. Singularity of $e^{z^{-1}}$ at $z = 0$ is of the type
60. $\text{Res}(e^{1/z})_{z=0} = \dots$ 61. Taylor's series expansion of $\sin z$ about $z = \pi/4$ is
62. Image of $|z| = 2$ under $w = z + 3 + 2i$ is 63. The poles of $\cot z$ are
64. If a is simple pole, then $\text{Res}[\phi(z)/\psi(z)]_{z=a} = \dots$
65. Bilinear transformation always transforms circles into
66. If $f(z)$ and $\bar{f}(\bar{z})$ are analytic functions, then $f(z)$ is constant. (True or False) (Mumbai, 2006)
67. The function $u(x, y) = 2xy + 3xy^2 - 2y^3$ is a harmonic functions. (True or False) (P.T.U., 2009 S)
68. The function $e^x \cos y$ is harmonic. (True or False)

69. $\int_C \frac{f(z)}{z-a} dz = 2\pi i f(a)$, if $z=a$ is a point within C . (True or False)
70. The transformation affected by an analytic function $w=f(z)$ is conformal at every point of the z -plane where $f'(z) \neq 0$. (True or False)
71. The function $\bar{z}$ is not analytic at any point. (True or False)
72. Under the transformation $w=1/z$, circle $x^2+y^2-6x=0$ transforms into a straight line in the w -plane. (True or False)
73. If $w=f(z)$ is analytic, then $\frac{dw}{dz} = -i \frac{\partial w}{\partial y}$. (True or False)
74. An analytic function with constant imaginary part is constant. (True or False)
75. If $u+iv$ is analytic, then $v-iu$ is also analytic. (True or False)
76. $f(z) = I_m(z)$ is not analytic. (True or False)
77. The cross-ratio of four points is not invariant under bilinear transformation. (True or False)
78. $z=0$ is not a critical point of the mapping $w=z^2$. (True or False)
79. $f(z) = \operatorname{Re}(z^2)$ is analytic. (True or False)
80. An analytic function with constant modulus is constant. (True or False)
81. The function $|\bar{z}|^2$ is not analytic at any point. (True or False)
82. If $f(z)=z^2$, then the family of curves $x^2-y^2=C_1$, and $xy=C_2$ are orthogonal. (True or False)